

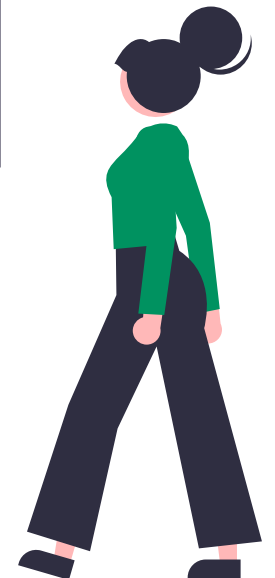
If you're here early...

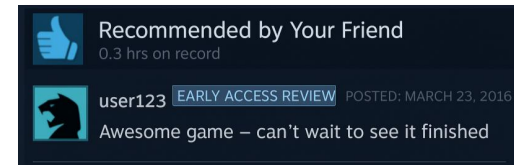
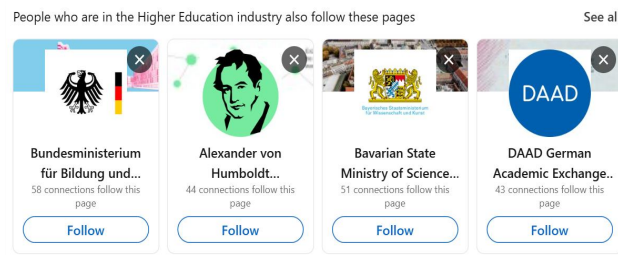
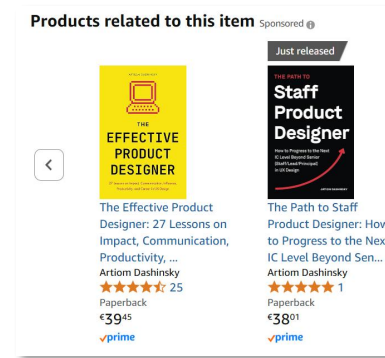
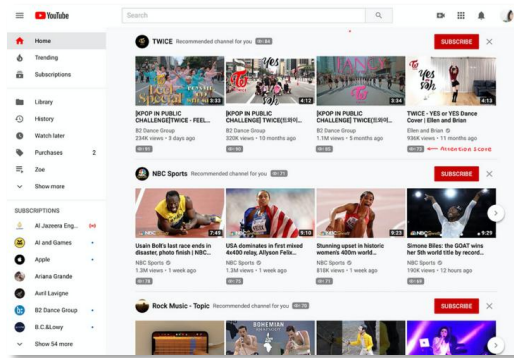
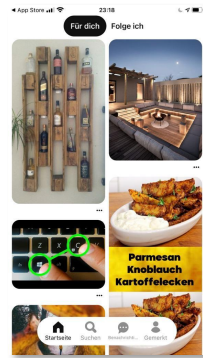
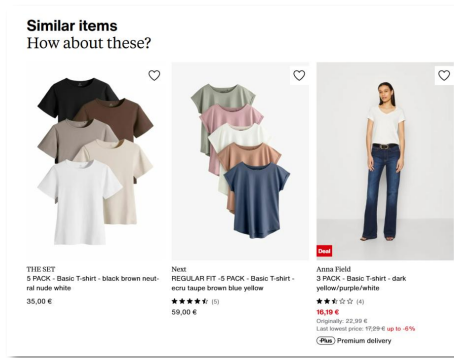
Prepare a pen + paper for our design activity during this session

Pick a use-case or domain for which you'd like to design a recommendation UI and write it down.

For inspiration, think about:

- **What** is recommended?
E.g. music, books, movies, fashion, jobs, locations, houses / flats, cars, restaurants, recipes, events, activities, exercises, learning material, courses, hiking trails, ...
- **Who** is the target group?
E.g. consider age, location, profession, skills, background, social context, personal interests and goals, ...
- **Device:** Mobile or desktop UI?





„5/5 – would join again!“

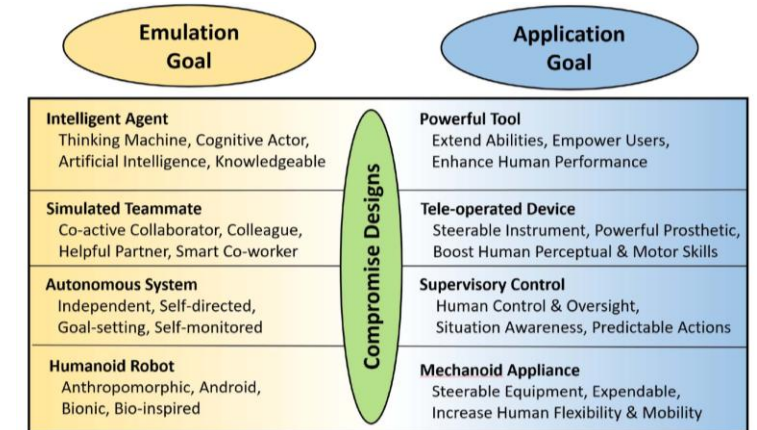
UIs & Interaction with Recommendations

Example using photos/material by Albrecht Schmidt

Recap & Last Quiz

Lecture Quiz

Explain the **two goals of AI**, and for each give an example of an interactive system that relates to that goal. Also describe which ML/AI tasks appear in your examples.



Emulate human abilities to understand them and build systems that perform tasks as well as or better than humans.

Apply AI methods to build usable products and services, to enhance the abilities of humans, improve people's lives...

Note: A specific technical method by itself does not have to imply a specific goal!

E.g. just because sth uses a neural network, it does not have to mean that it's e.g. emulation. It also depends on what the method is used for, and what the interaction design is (e.g. both a humanoid robot and a "fix grammar" tool might use an LLM in the background).

Lecture Quiz

List three AI methods and give examples of what you could do with them for UIs.

Generative models,
e.g. LLM: “useful for contextually adjusting input material and could be used to provide summaries of longform content.”

Clustering: “Make folders of the same things”

Reinforcement learning: “An UI could use this to gradually optimize the layout or suggestions it shows you based on what you actually click on over time.”

Natural Language Processing: “to understand language input, e.g. for a chatbot”

Recommendation system: “Show relevant items”

Classification for adaptation: “Is the phone near the ear when taking a call the sound is played quieter. Is the phone held in front of the face it is immediately on speaker.”

Lecture Quiz

Reflect on software, apps, devices that you use:
Which of their features might use classification, regression, clustering, ...?

“**Segments at the YouTube start page** that all contain similar content (**clustering, classification**)”

Classification: “my **mail** accounts use classification to **detect spam**”

Regression: The **touchscreen keyboard** adapts to my typing behaviour by predicting more accurate touch positions, improving typing accuracy over time.

“In **Google Maps**, the **arrival time** prediction uses **regression** because it predicts a continuous value, like minutes.”

Clustering: “My **photo gallery** recognizes faces and tries to classify them as specific people. Then it clusters these into individual photo albums.”

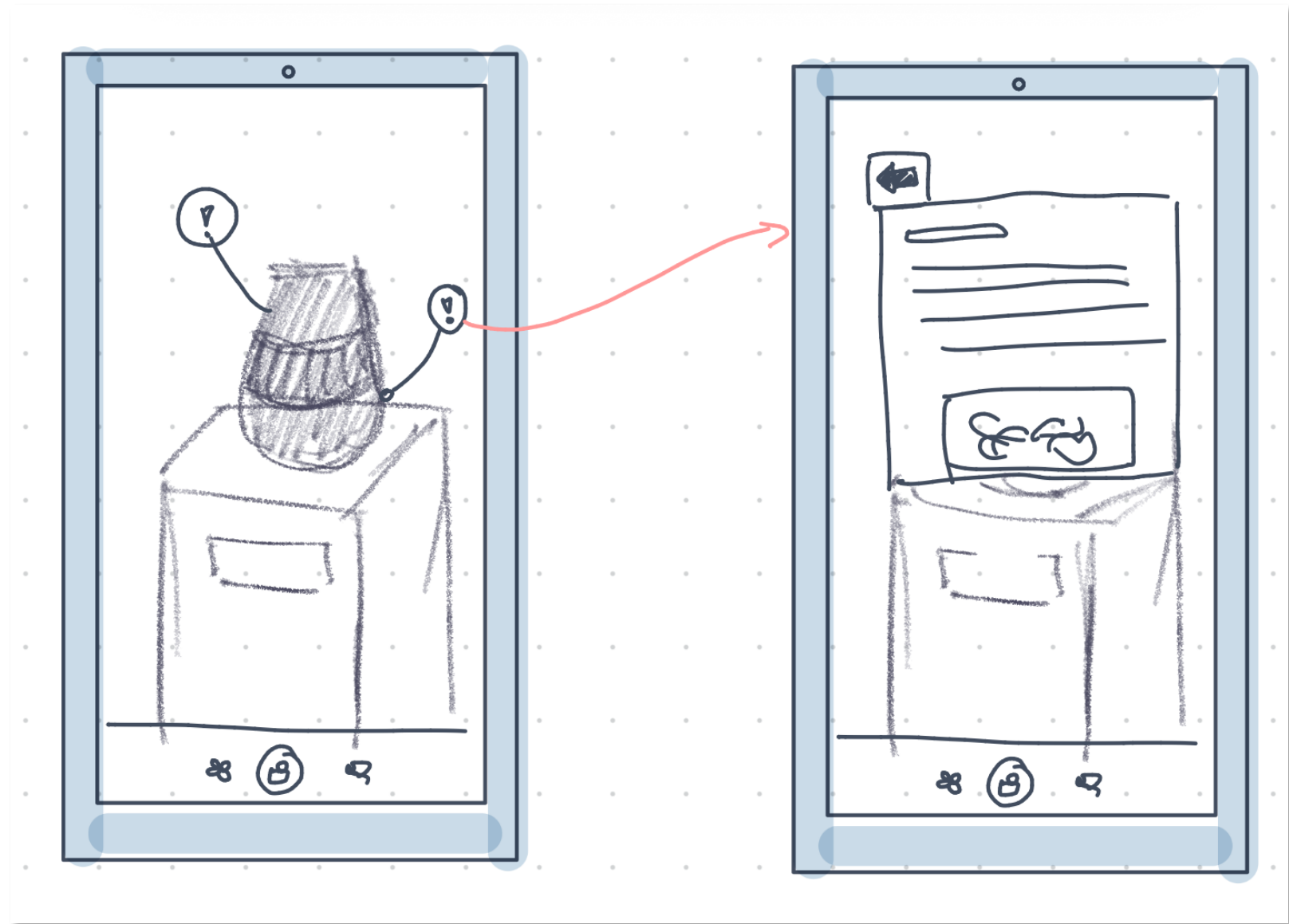
“**Recommendation systems:** **personalize the homepage** of Netflix, YouTube, or a news app.”

Lecture Quiz

Intelligent features for...

Build an **app for a museum** that shows infos about the object in the camera view.

The App need visual and informational **data about all the objects** in the museum and it needs the **live camera feed**. It needs to be able to **recognize similarities** between the data objects from the museum and the objects shown in the users camera.



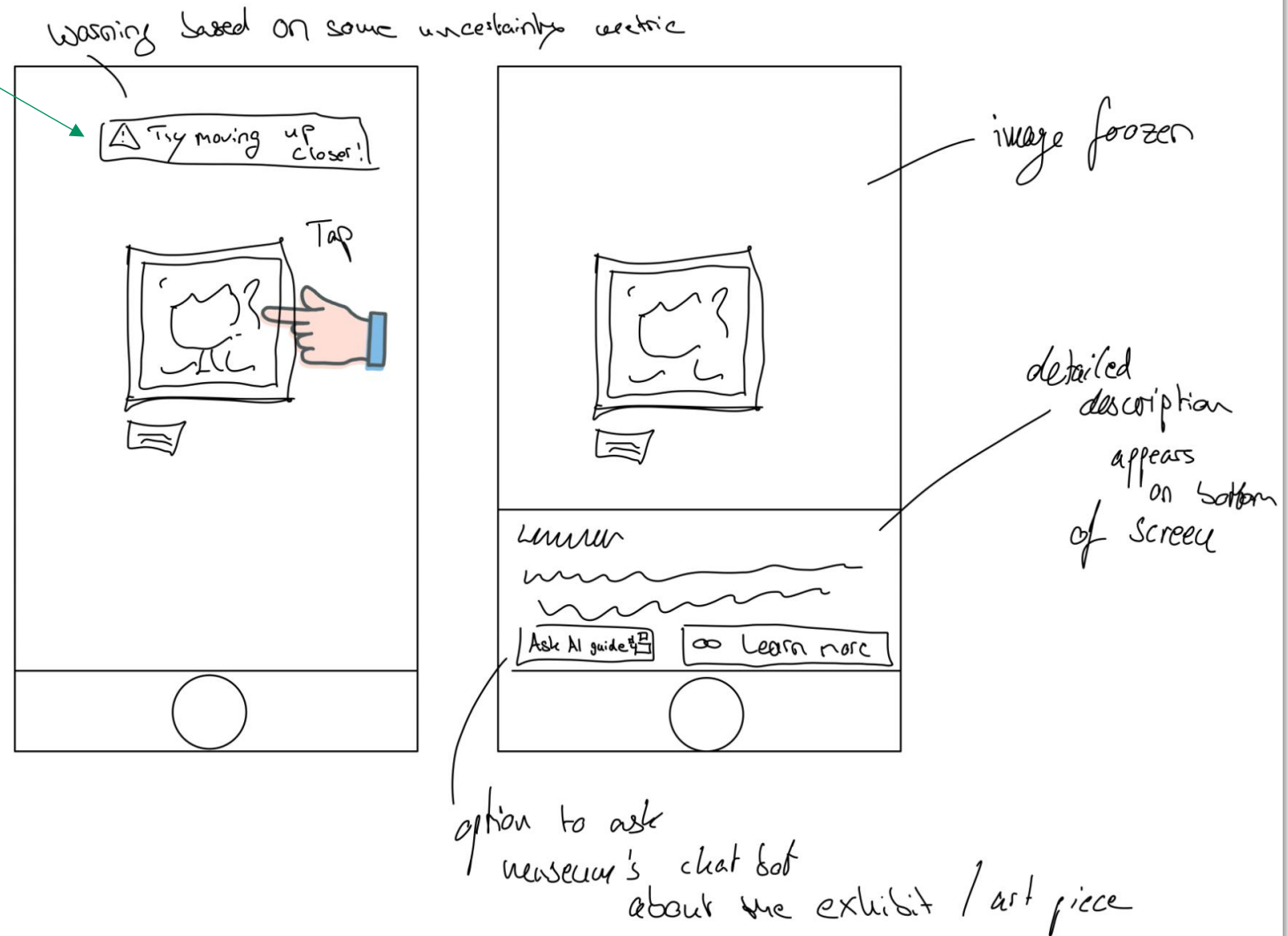
Lecture Quiz

Intelligent features for...

Build an **app for a museum** that shows infos about the object in the camera view.

The AI would be used to detect what item of the exhibit is currently captured in the live camera view and would allow users to tap the item to reveal additional information. The required data for this would be multiple images from multiple angles of the exhibit to make sure the detection works well and doesn't confuse different items. There could also be a button to "Ask digital tour guide" where you could then chat with an AI chatbot that knows even more about the exhibit than is possible to be displayed in the short detail-popup.

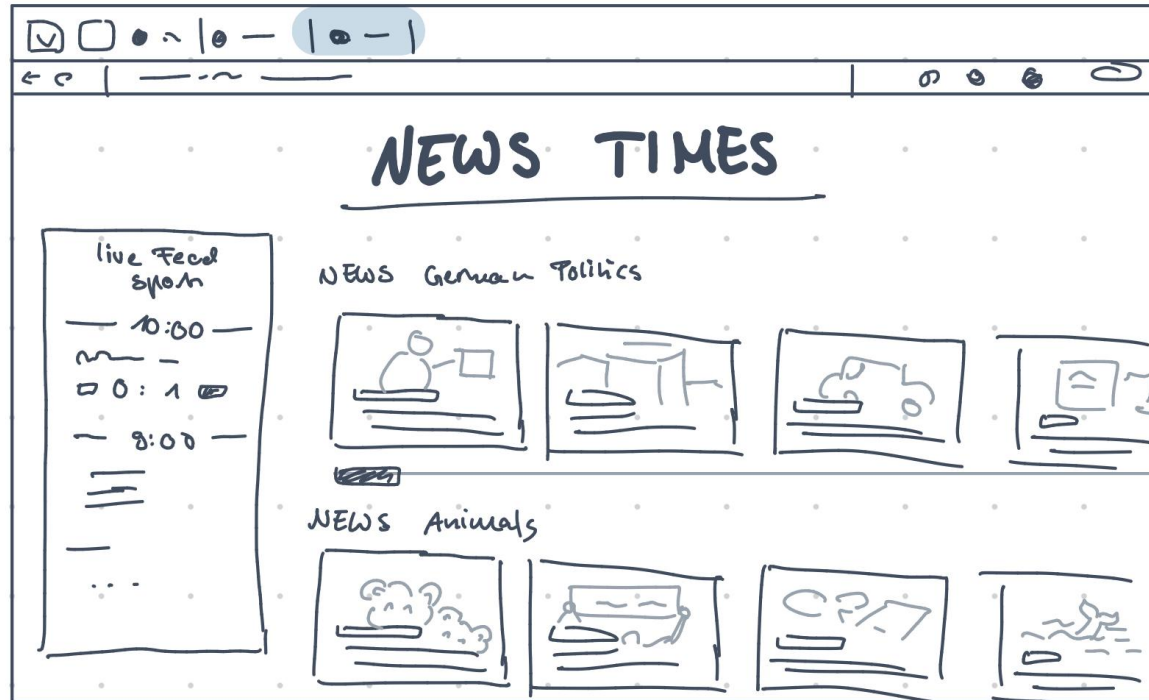
Guiding the user's input



Lecture Quiz

Intelligent features for...

Improve the layout of a **news website** based on user interests.



Explicit user feedback

News Aggregator

Recommended

Four word headline, surely

Date

Logline that's as long as you'll need to know if you want to read this.

Not Interested

Four word headline, surely

Date

Logline that's as long as you'll need to know if you want to read this.

Not Interested

Four

Date

Logline that's as long as you'll need to know if you want to read this.

Not Interested

Trending

Four word headline, surely

Date

Logline that's as long as you'll need to know if you want to read this.

Not Interested

Four word headline, surely

Date

Logline that's as long as you'll need to know if you want to read this.

Not Interested

Four

Date

Logline that's as long as you'll need to know if you want to read this.

Not Interested

Under-reported

Four word headline, surely

Date

Logline that's as long as you'll need to know if you want to read this.

Not Interested

Four word headline, surely

Date

Logline that's as long as you'll need to know if you want to read this.

Not Interested

Four

Date

Logline that's as long as you'll need to know if you want to read this.

Not Interested

The System needs to collect data about the type of news the user **often clicks on or stays there the longest**. Clustering could be used to group things of similar interest to the user.

Implicit feedback

Lecture Quiz

Intelligent features for...

Enabling users to define their **own gesture interactions** for smart home devices

Shows the currently performed user gesture to record for this command (camera feed)

Gesture stored in a database, to be compared to future input to trigger the right command

Shows the command for which to define a new gesture

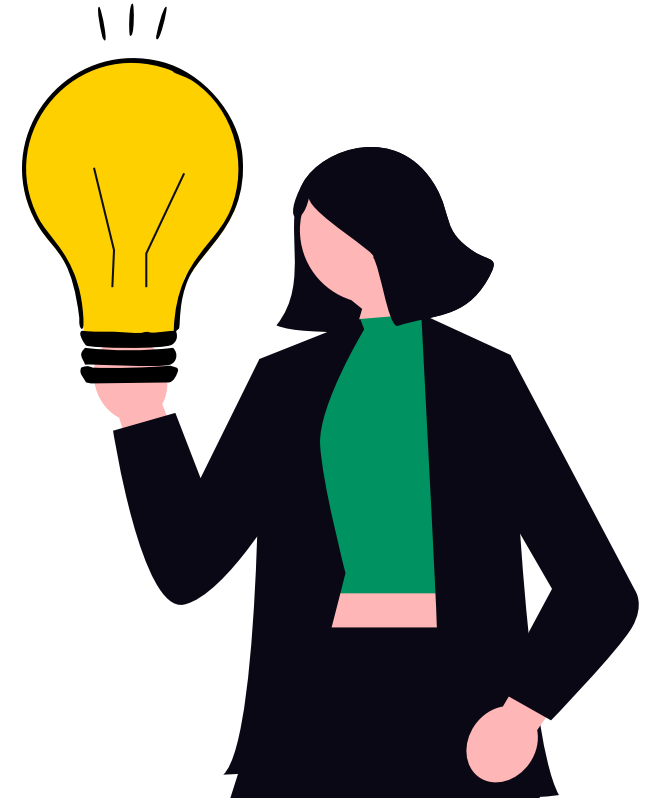


This lecture

Learning goals

After this session, you should be able to:

- ✓ **Explain** the motivation of recommendation features in interactive systems in your own words with concrete examples
- ✓ **Explain** the differences between fundamental approaches for computing recommendations, incl. from a UI & interaction design perspective
- ✓ **Create** a UI design concept / sketch for a recommendation-based feature that handles common requirements (e.g. addresses the “cold start” problem)
- ✓ Using the lecture notebook: **Implement** a simple recommendation system in Python



Intro

Why think about recommendations in UIs?

Pragmatic motivation

Recommendations are very common in end-user applications and sometimes pretty much the „whole“ UI!

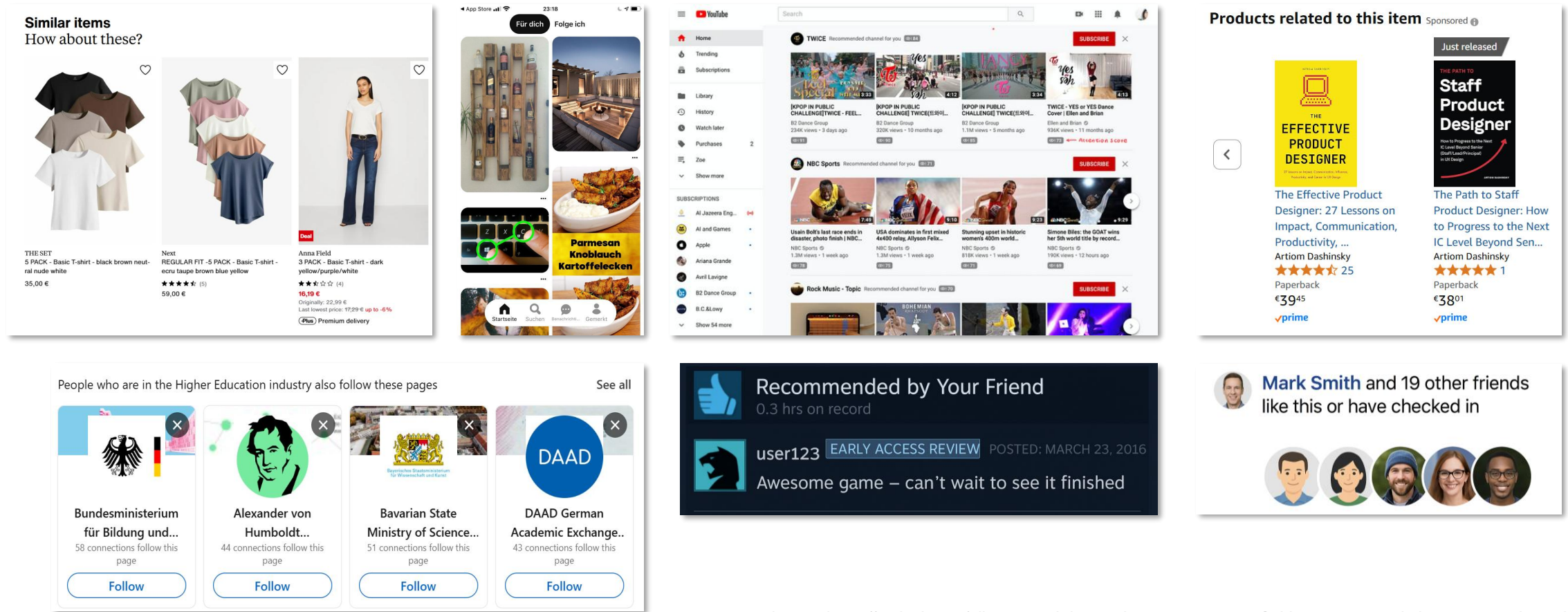
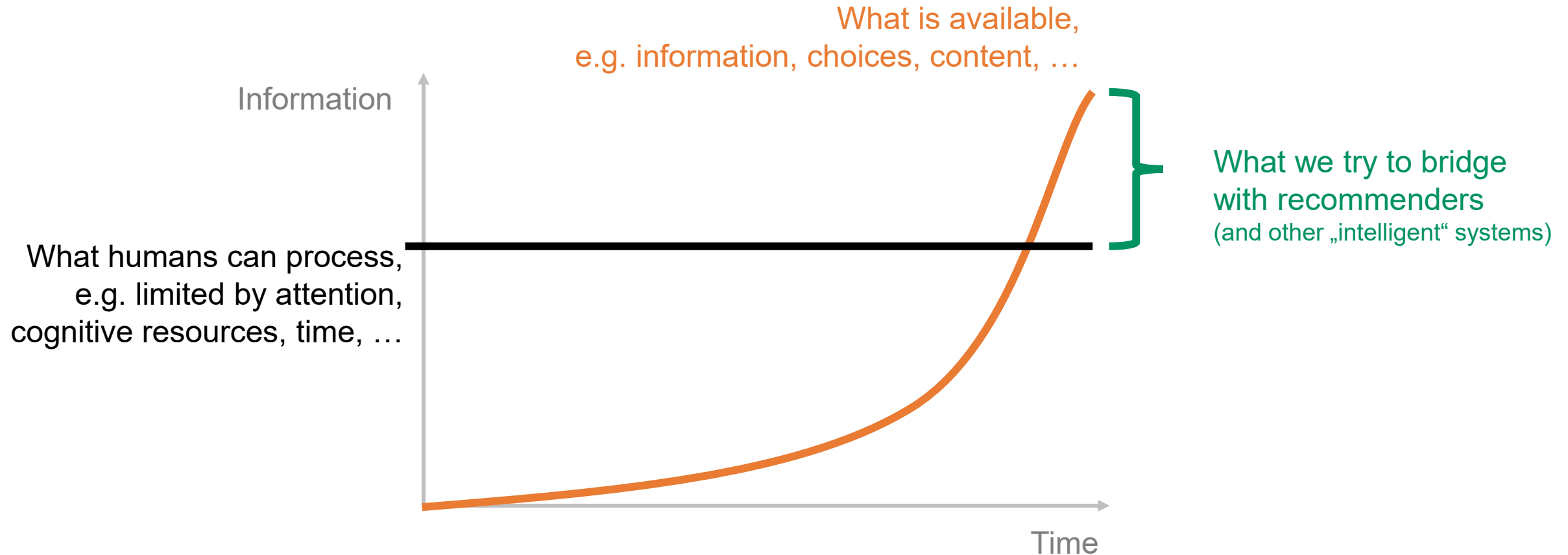


Image: <https://uxdesign.cc/all-eyes-on-it-increasing-transparency-of-video-recommendations-on-youtube-ca008d68d816>

Conceptual motivation

Information overload and supporting people in decision-making



Example beyond everyday use-cases

Recommendations for diversion decision-making for pilots

1

2

3

4

Diversion = sth unexpected has happened and you need to land the plane at a different airport than planned – but which one?

- 1) Recommendations
- 2) Edit recommendation criteria
- 3) Local hints during normal flight
- 4) Adjusted local hints during emergency

[Zhang et al., 2024]

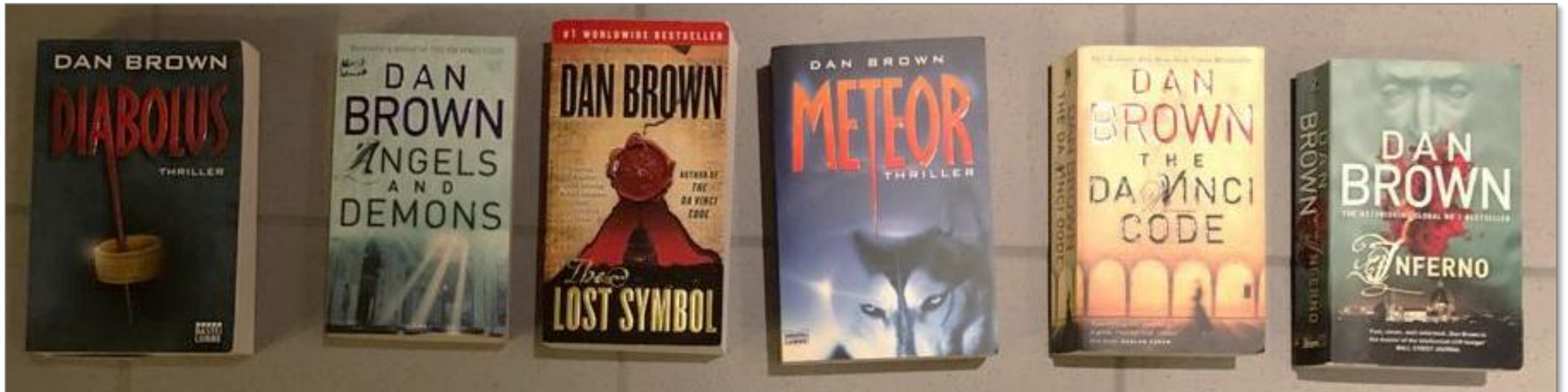
Computing Recommendations

The algorithmic part: How can we compute recommendations?

Human recommendations

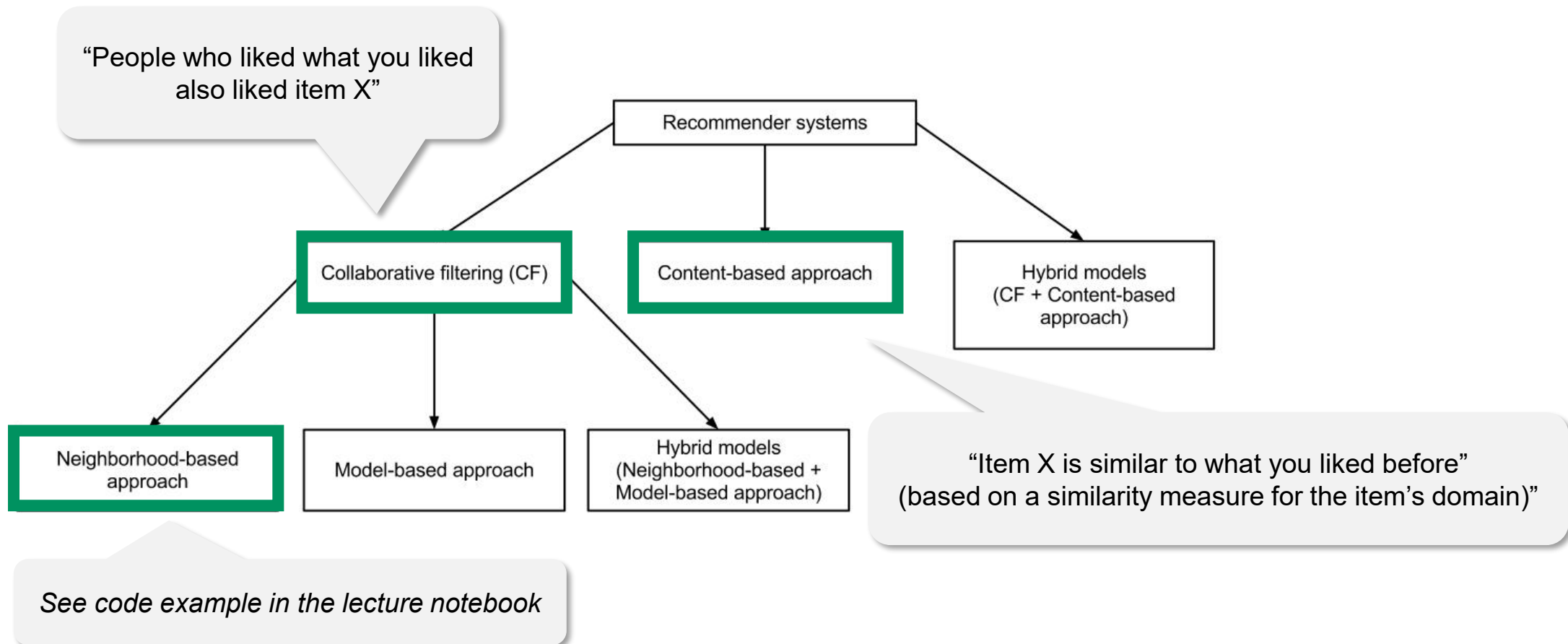
Which books would you recommend to my colleague? 😊

Some of the books he read:



Overview

Of typical approaches



https://en.wikipedia.org/wiki/Collaborative_filtering

Content-based filtering

Basic idea

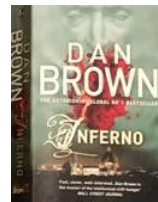
1. Determine what's relevant to **describe items**

→ List of features

Language,
genre,
time,
...

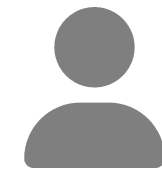
2. Determine feature **values** for all items

→ Items as numerical vectors



English,
thriller,
present,
...

$\begin{pmatrix} \dots \\ \dots \\ \dots \end{pmatrix}$

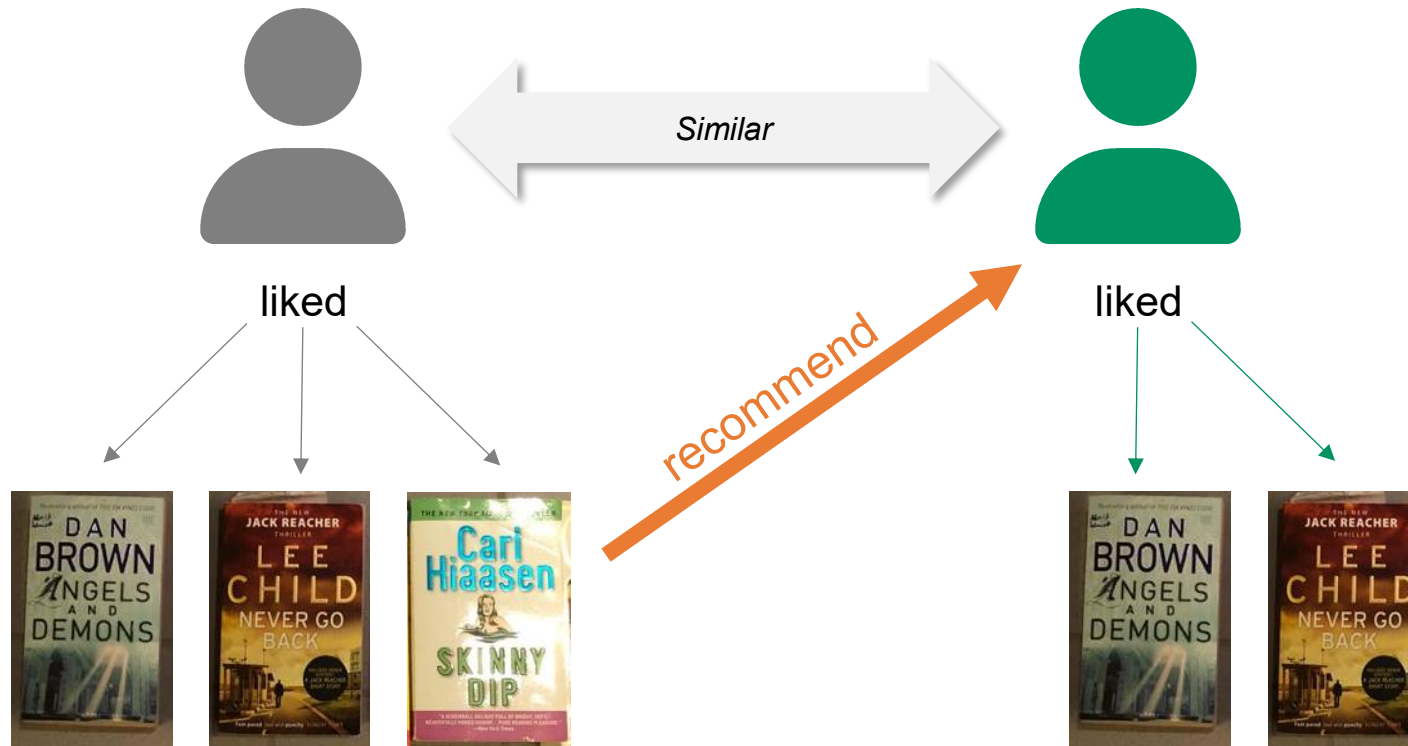
 $\begin{pmatrix} \dots \\ \dots \\ \dots \end{pmatrix}$

score = ...

4. **Recommend** items that fit to user, calculated with item & user vectors
(e.g. cosine similarity / angle between the vectors)

User-based Collaborative Filtering

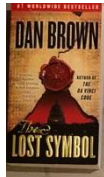
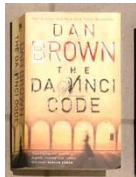
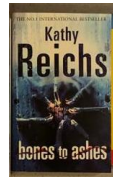
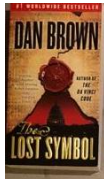
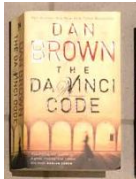
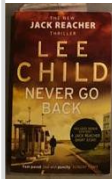
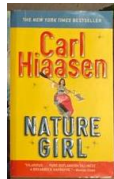
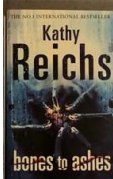
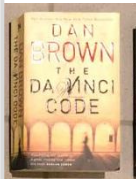
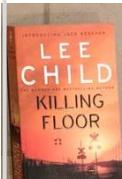
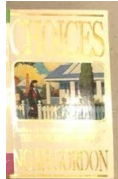
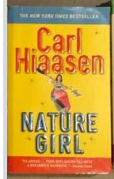
Basic idea

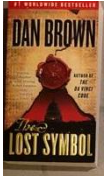
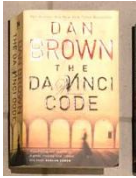
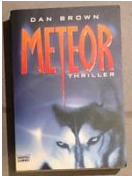
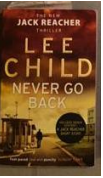
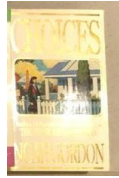
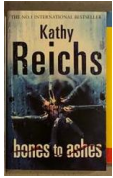


Example based on <https://medium.com/swlh/how-to-build-simple-recommender-systems-in-python-647e5bcd78bd>

Illustrated computation example

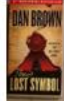
Note: Many algorithms and variants exist,
consider this as one possible example (for CF),
not a specific implementation

												
	8	7	9	4	9		10			4		7
		2				6	9					
	4		6	10	8		9		6	7	9	8
	4	?	6	?	?	?	?	9	5	?	10	?

Example: Similarity matrix

Values just for illustration (not actually calculated)

												
	8	7	9	4	9		10			4		7
		2				6	9					
	4		6	10	8		9		6	7	9	8
	4	?	6	?	?	?	?	9	5	?	10	?

Similarity matrix (part)

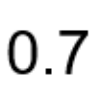
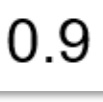


Example: Weighted rating matrix

Rating matrix
(part)

		
	7	4
	2	
		10
	?	?

Similarity matrix
(part)

	 0.4
	 0.7
	 0.9

Weighted rating matrix
(part)

		
	2.8	1.6
	1.4	
		9.0
	?	?

Recommendation

Weighted rating matrix
(part)

		
	2.8	1.6
	1.4	
		9.0
	?	?

Sum up ratings per book

		
 	4.2	
 		10.6

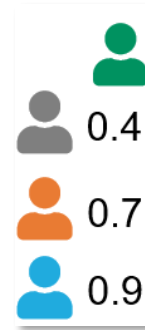
/

$$0.4 + 0.7 = 3.8$$

/

$$0.4 + 0.9 = 8.2$$

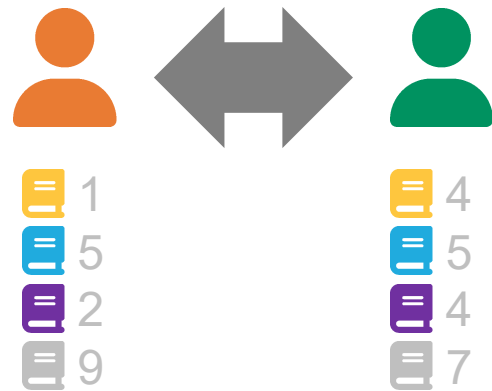
Sum up similarities
for involved users



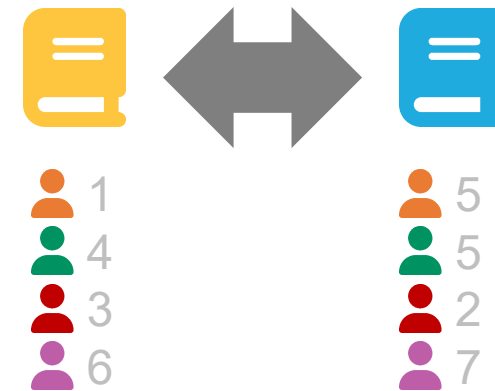
Similarity matrix



Two flavours of Collaborative Filtering



Across both variants, a key aspect of CF is that we **don't need any information from the actual domain** (e.g. book content) to assess similarity!



User-based (prev. slides)

- Calculate **similarity between users** based on their item ratings
- Suggest items that similar users liked

Item-based

- Calculate **similarity between items** based on user ratings for those items
- Suggest items similar to items the user already liked

UI Design

How are recommendations presented in the UI?

Activity: „Design along“

Design a UI with recommendations

Pick a use-case or domain and write it down.

For inspiration, think about:

- **What** is recommended?
E.g. music, books, movies, fashion, jobs, locations, houses / flats, cars, restaurants, recipes, events, activities, exercises, learning material, courses, hiking trails, ...
- **Who** is the target group?
E.g. consider age, location, profession, skills, background, social context, personal interests and goals, ...
- **Device:** Mobile or desktop UI?



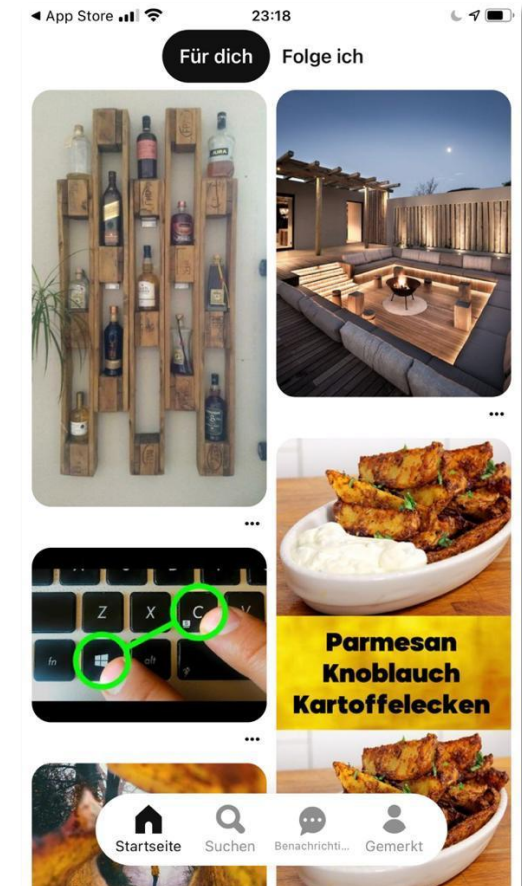
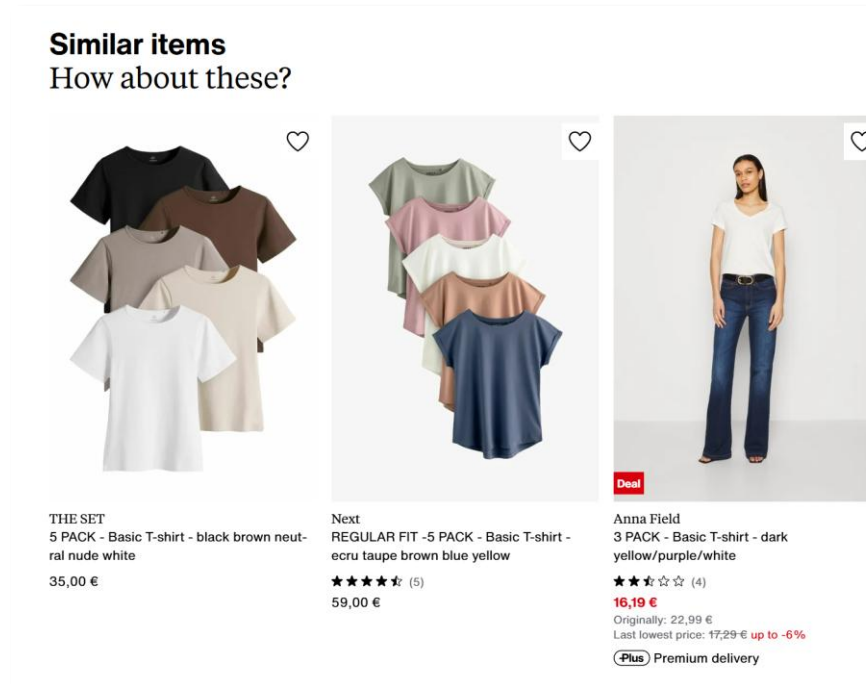
Presentation

Is a recommended item actually presented as a „recommendation“ in the UI?

Some UIs **explicitly** say „recommendation“:

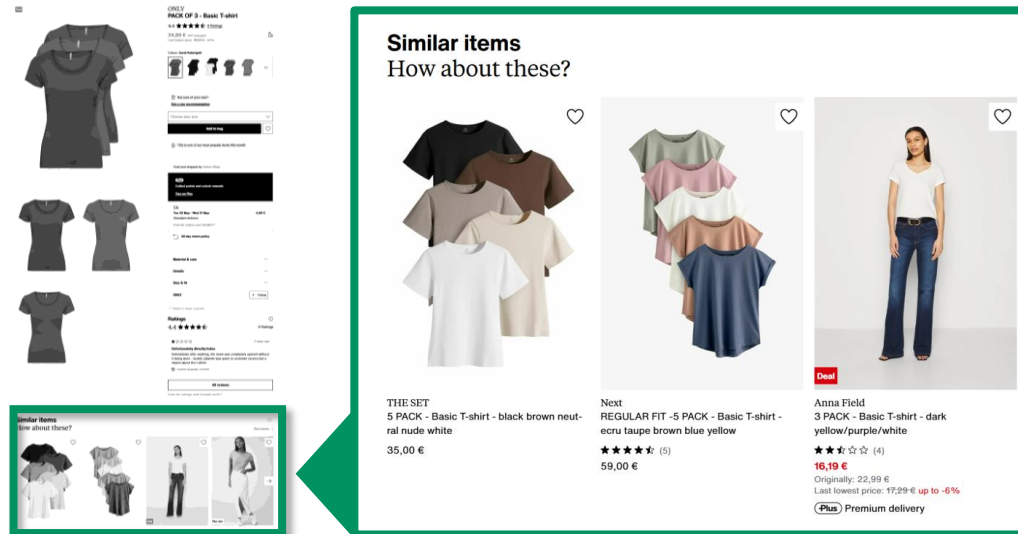
View all recommendations →

What about these UIs: Where are the **indicators** for „recommendation“ here?



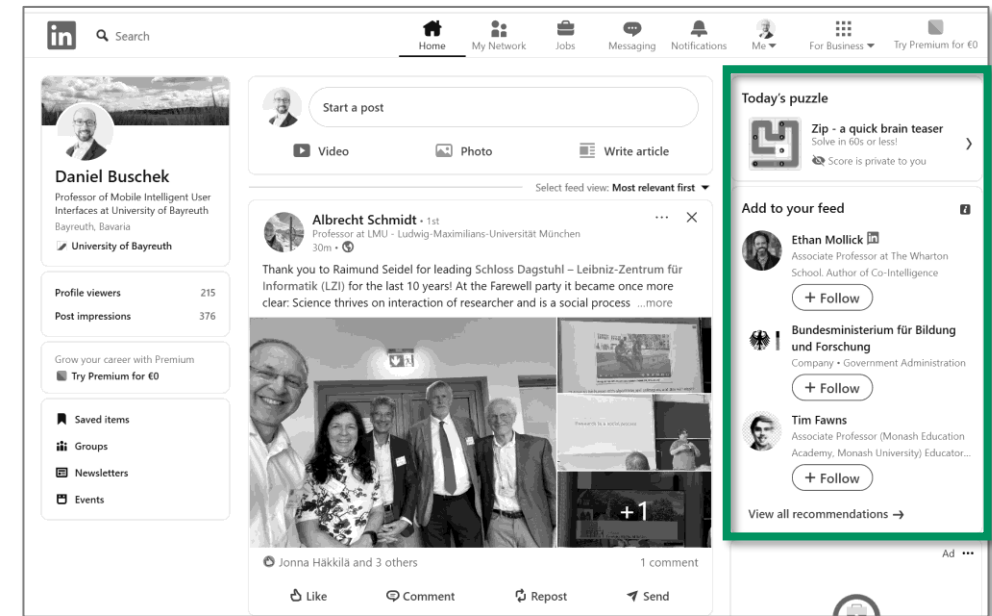
Integration

How to integrate recommendations into the overall UI?



In main view

E.g. shopping UIs: “similar” items directly below current item in the main view

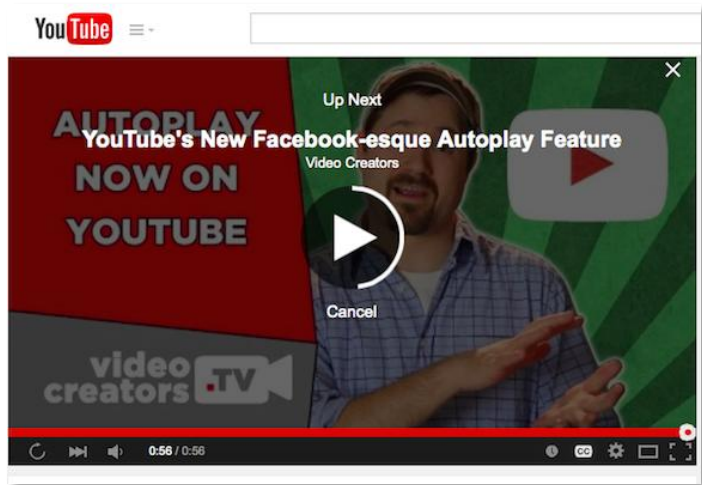


As secondary / context element

E.g. as a sidebar, next to the main view / main task

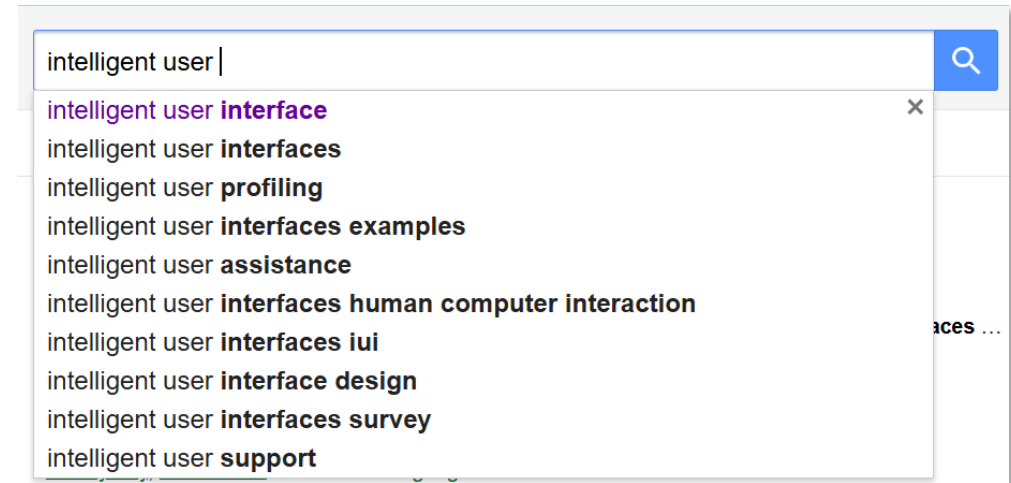
Integration

How to integrate recommendations into the overall UI?



Temporal integration

E.g. recommended video plays directly after the current one, in the same UI view



Task integration

Directly part of the main task;
e.g. recommended search terms

Note: We'll cover this "suggestion"-style task integration more in a separate session on text entry.

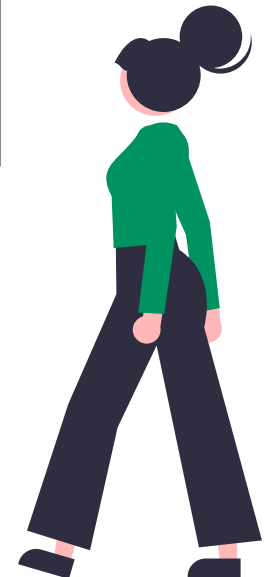
Activity

Design a UI with recommendations

Make a **first sketch** of a UI for your use-case.

Think about:

- **Presentation:** Is a recommended item actually presented as a „recommendation“ in the UI?
- **Integration → layout:**
 - How are recommendations integrated in the UI, e.g. in the main view vs a sidebar? Appearing over time?
 - Are the recommendations the main point of the app or do they support another task?



Quantity

How many recommendations to display in the UI?



Too few

- Risks missing relevant options
- User might feel constrained, or like not having a real choice

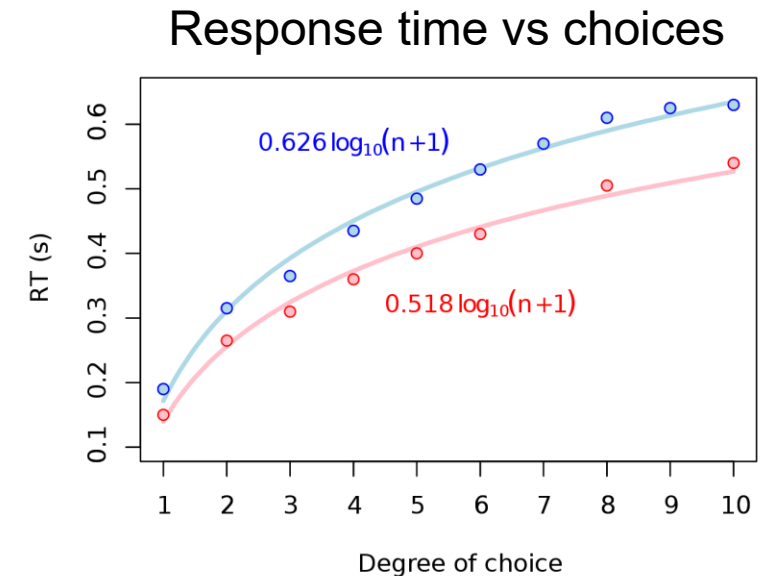
Too many

- Potentially overwhelming
- Increases decision time (see next slide)
- Might take UI space from other features

Quantity

How many recommendations to display in the UI?

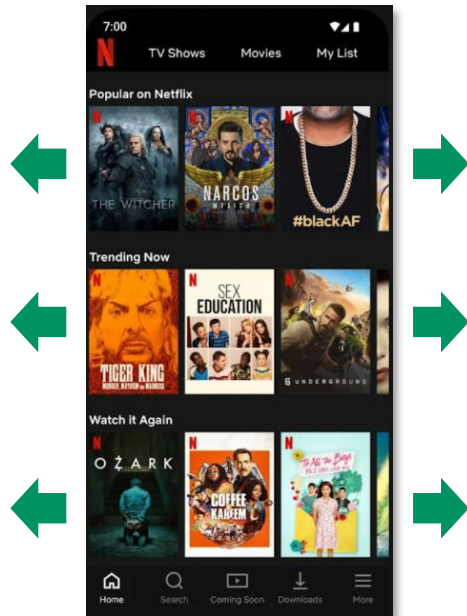
- Not just „as many as possible“ given the UI space!
- Also consider:
 - Task and goals (e.g. is making a decision the main task?)
 - Context (e.g. at desk vs in the car)
 - Cognitive resources (e.g. main task demands, distraction)
- **Hick's Law:** „*increasing the number of choices will increase the decision time logarithmically*“



https://en.wikipedia.org/wiki/Hick%27s_law

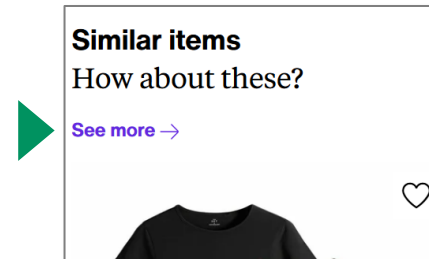
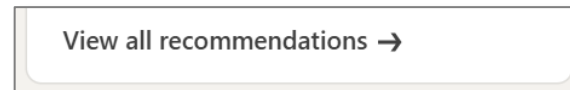
Quantity

Some UI design options to indicate and accommodate “more” recommendations



Carousels

Include half-cut content to indicate there's more!



“More” buttons

→ user can explicitly request more



Endless scroll

→ user implicitly triggers more

Image: <https://www.everys.com/articles/infinite-scrolling-pages-and-how-google-crawls-them>

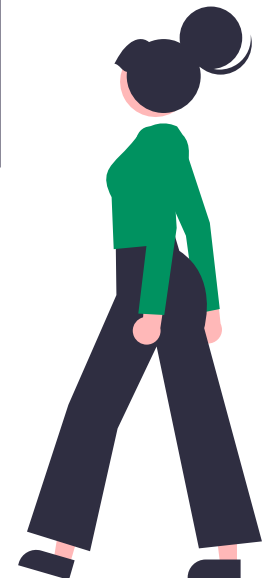
Activity

Design a UI with recommendations

Adjust your UI sketch or sketch a new iteration.

Think about **quantity**:

- How many recommendations does your UI show at a time?
- You could think of your target group and their goals to help make this design decision



Detail

Which & how much information to show for each recommendation?



Too little info

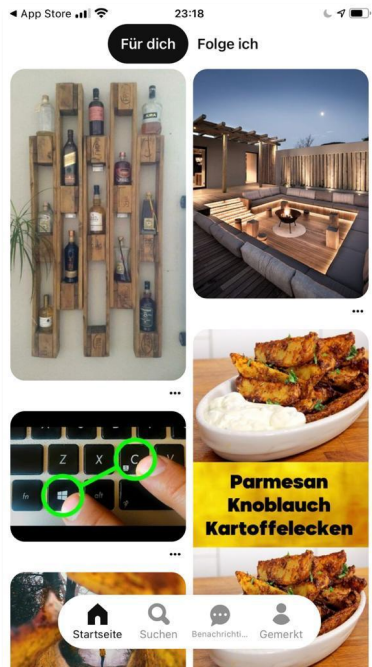
- Risks missing the key differences between the options
- User might not feel able to make a well-founded decision

Too much info

- Potentially overwhelming
- Increases reading / decision time
- Might take UI space from other features

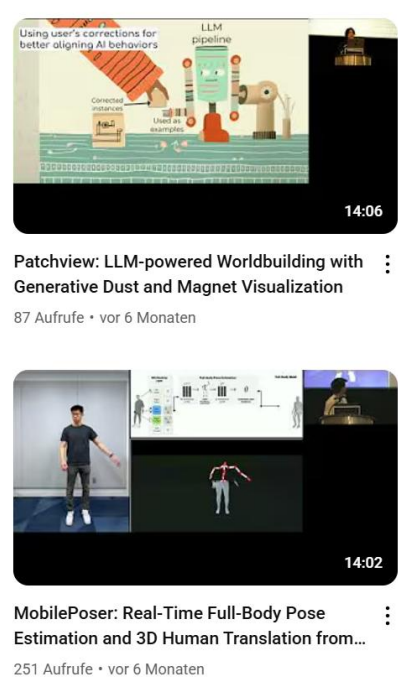
Detail

Typical UI design choices for everyday applications



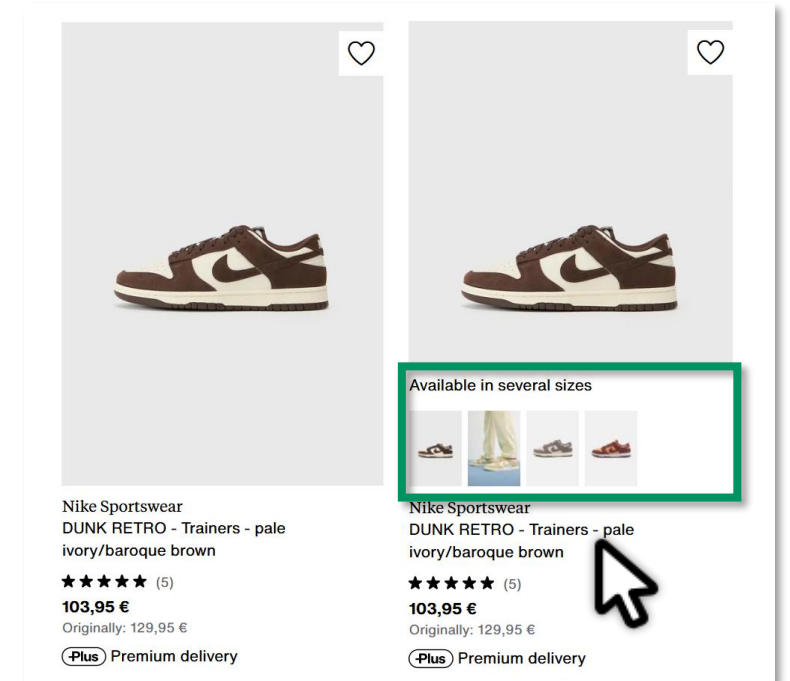
Key visual only

Typical for images, movies, anything with a thumbnail



Visual + text

e.g. thumbnail + title on YouTube



Details on demand

→ user can request more details through interaction (e.g. hover)

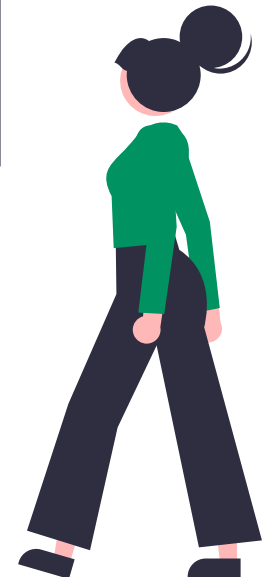
Activity

Design a UI with recommendations

Adjust your UI sketch or sketch a new iteration.

Think about **detail**:

- Which & how much information does your UI show for each recommendation?
- You could think of your target group and their goals to help make this design decision

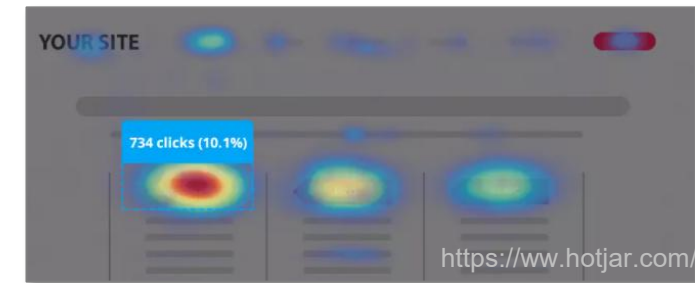
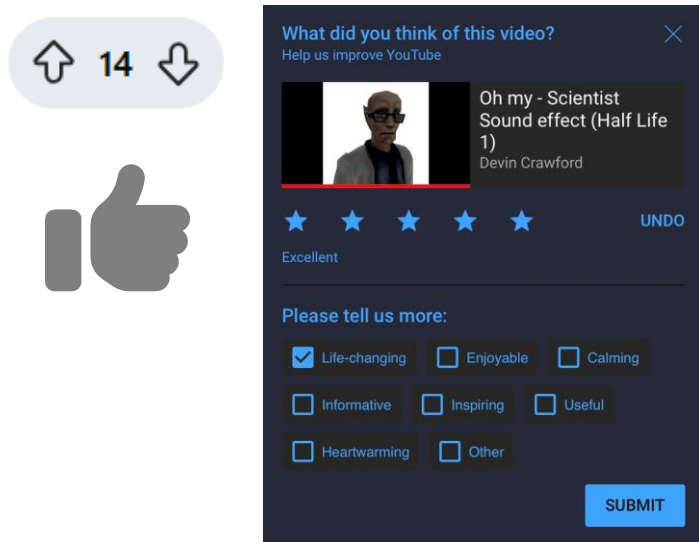


Interaction Design

How can users interact with recommendation-related UI elements?

Feedback interactions

How do users give feedback on recommendations?



Explicit feedback

- Giving feedback to the system *is* the main reason for these interactions
- E.g. voting up/down, rating a recommendation

Other interactions

- Still explicit user actions, but not direct rec. feedback
- E.g. share, comment, bookmark, **review** (some reviews are feedback on a product, but not necessarily feedback on a recommendation)

Behaviour signals

- Feedback inferred from behaviour
- E.g. reading/viewing time, scrolling, mouse movements, revisits, eye-tracking, etc.

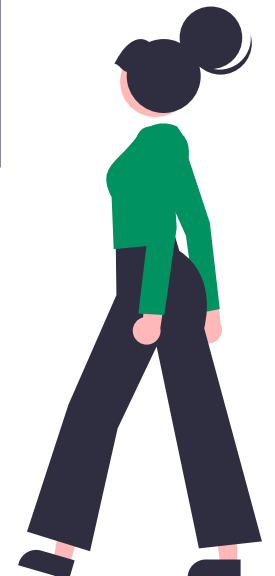
Activity

Design a UI with recommendations

Adjust your UI sketch or sketch a new iteration.

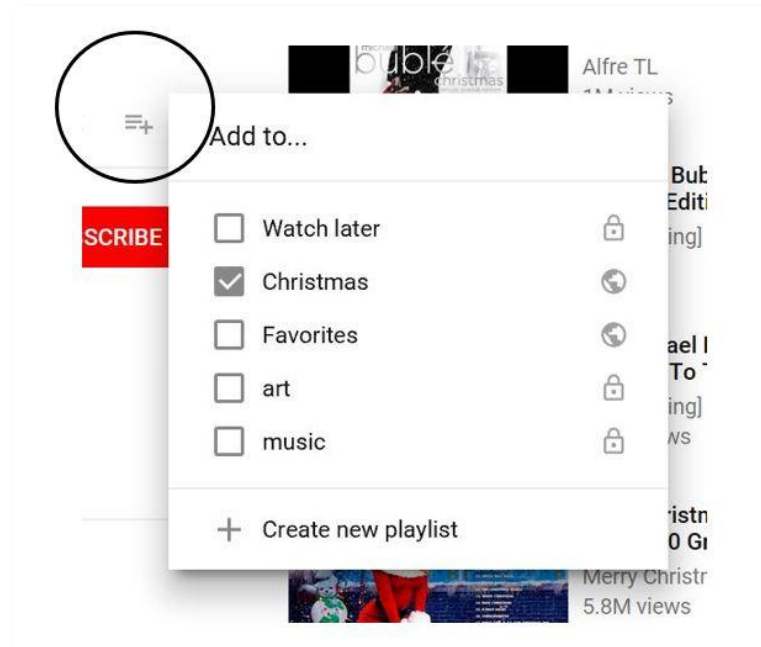
Think about **feedback interactions**:

- **Should users give feedback** on the shown recommendations in your UI?
- If so, **how** can they do so?
E.g. add UI elements for explicit feedback
or think about which interactions could be used
as implicit feedback



Functional interactions

How can users „work“ with recommendations and/or with the recommendation system?



Keep & export

- Interactions to **keep** and/or (later) **export** material, also **beyond the recommendation context**
- E.g. pin, bookmark, add to ..., share, copy, etc.

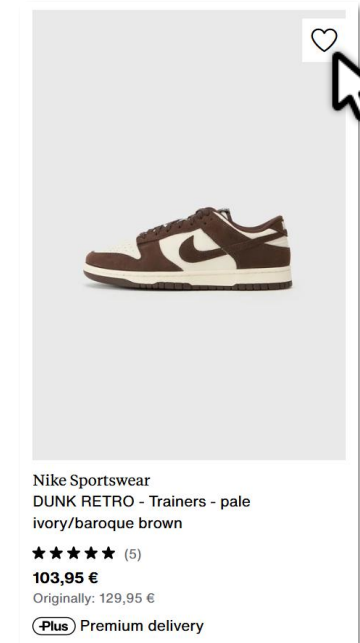


Image: <https://www.digitalunite.com/technology-guides/tv-video/youtube/how-create-playlist-youtube>

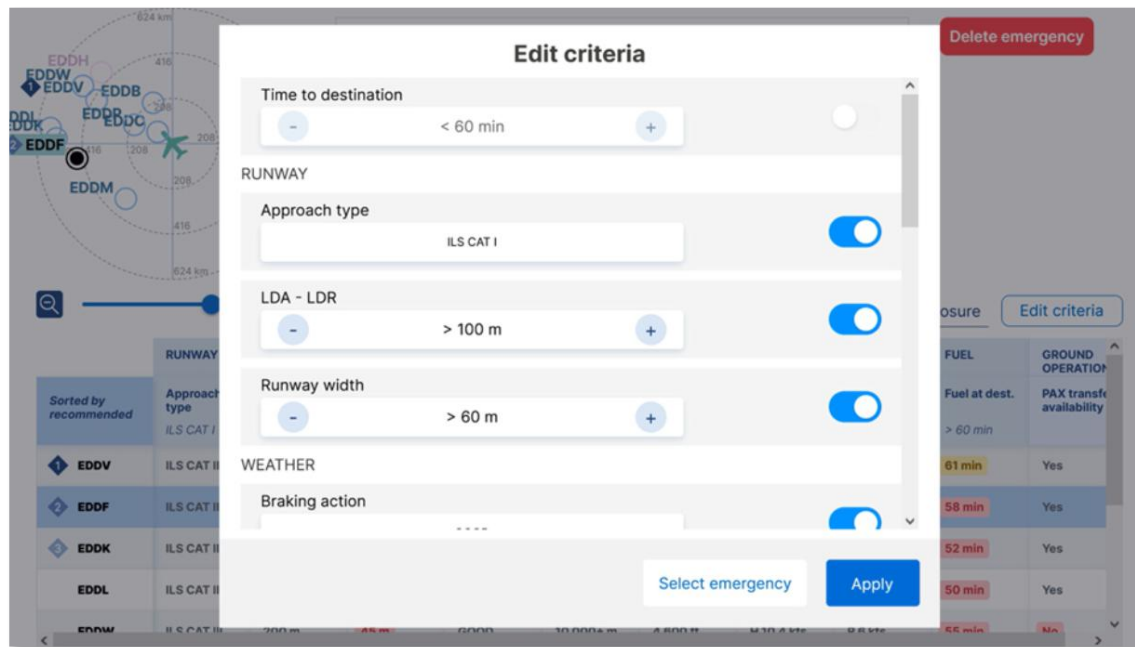
Functional interactions

How can users „work“ with recommendations and/or with the recommendation system?

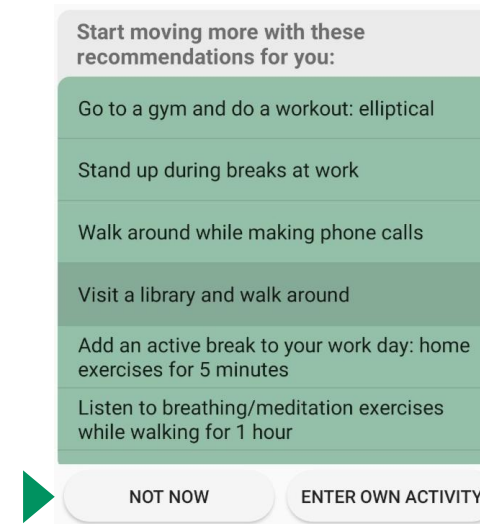
Manipulate

- Interaction to **work** with the material **within the recommendation context**
- E.g. dismiss, compare, re-rank, refine, change recommendation settings, use as new query

Diversion recommendations for pilots, with editable criteria



Customizable fitness app recommendations



Images: <https://dl.acm.org/doi/10.1145/3631700.3664867>, <https://www.freeletics.com/en/blog/posts/how-to-successfully-plan-fitness-throughout-the-week/>

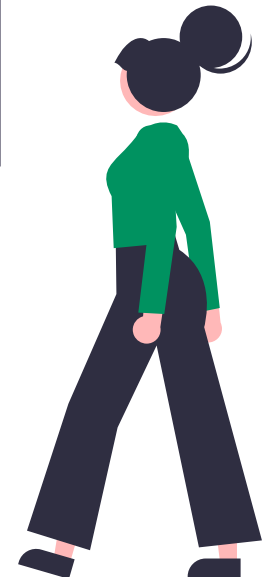
Activity

Design a UI with recommendations

Adjust your UI sketch or sketch a new iteration.

Think about **functional interactions**:

- Should users „work“ with recommendations?
- If so, how?
 - Within your app?
E.g. dismiss, compare, re-rank, refine, change recommendation settings, use as new query, ...
 - Beyond it?
E.g. pin, bookmark, add to ..., share, copy, ...



Algorithm-related UI Design

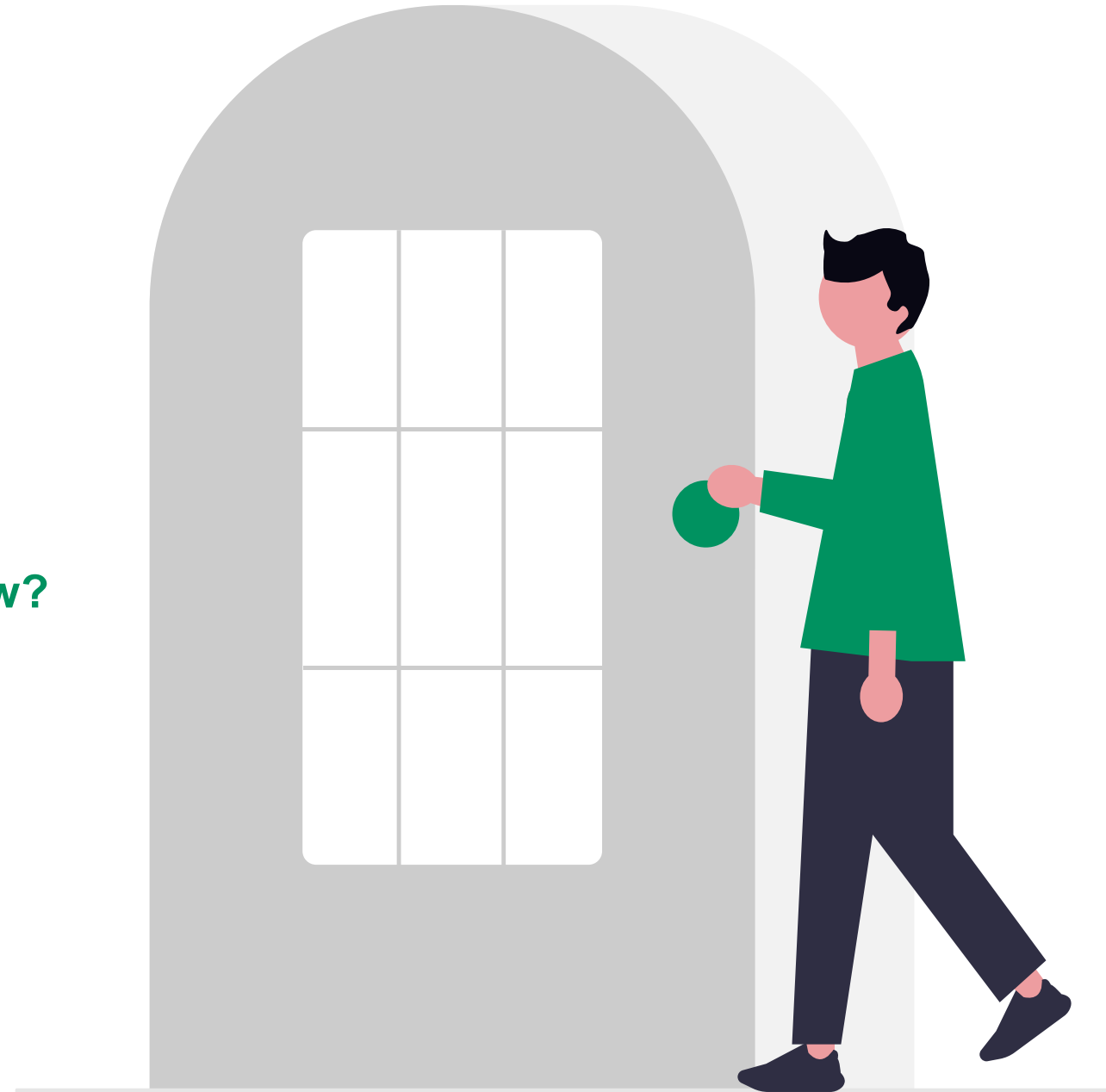
Bringing algorithm + UI together:

How do aspects of the algorithm impact the UI design and its requirements?

And how do aspects of the use-case impact our choice of algorithm/parameters?

What if a new user enters?

What can we recommend for this user? How?



The cold-start problem

Recommendations need initial information (e.g. user ratings)

New user
no ratings
or interactions yet



New item
not rated yet
by anyone



New system
just created, no info
on users/items

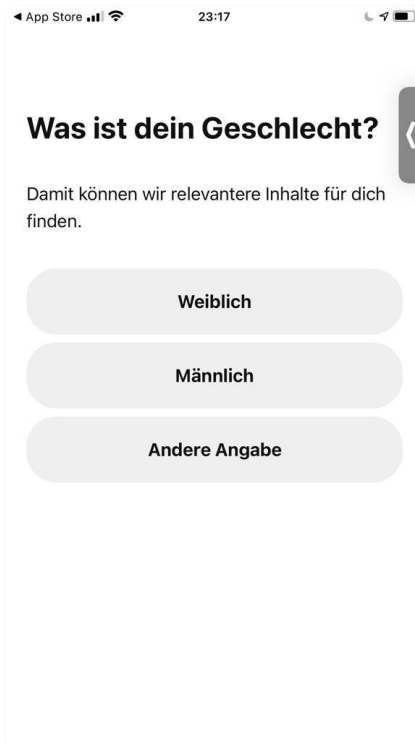


Solutions

- **Hybrid** approaches:
e.g. content-based filtering at the beginning,
then shift to collaborative filtering
- Use **background info** to get started
(social, demographics, context, ...)
- Require initial **user interaction**
(e.g. questionnaire, ask for examples, ...)

Onboarding UIs

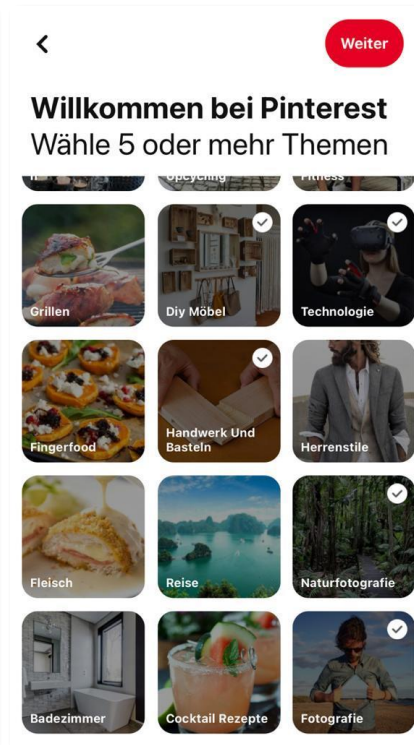
Example: Pinterest



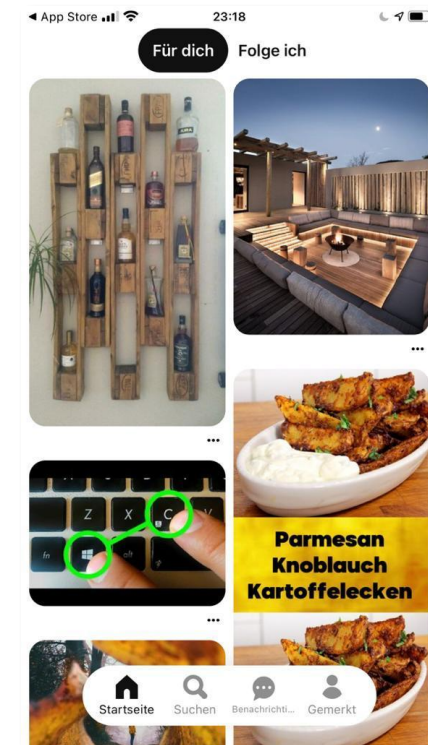
Demographics



Context



Explicit interest



First feed

based on this information; then collect more feedback via likes, interactions, ...

Onboarding UIs

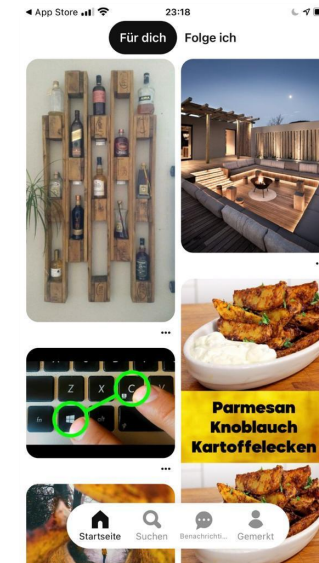
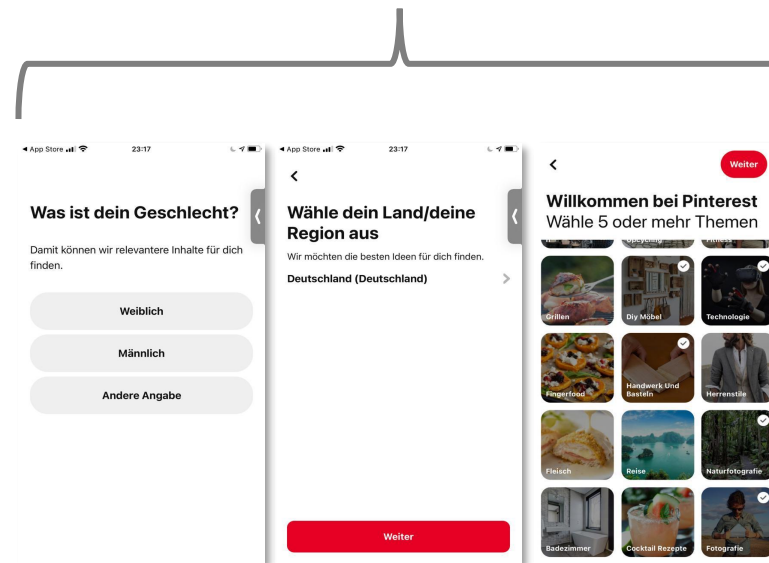
General design tradeoff for onboarding UIs

User **effort** and **delay**
before getting to the interesting
part of the application

VS

Quality of initial
recommendations

To improve this tradeoff, we could think of ideas to **make the onboarding experience better**, even if we ask for a lot of info.



Or we could think of ideas to **make the initial UI state feel better**, even if (some) recommendations aren't great yet.

Or we could think of ideas to **merge these two phases**, e.g. by integrating the onboarding into the main UI/interaction over time.

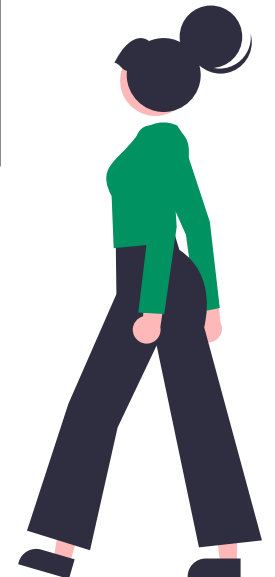
Activity

Design a UI with recommendations

Adjust your UI sketch or sketch a new iteration.

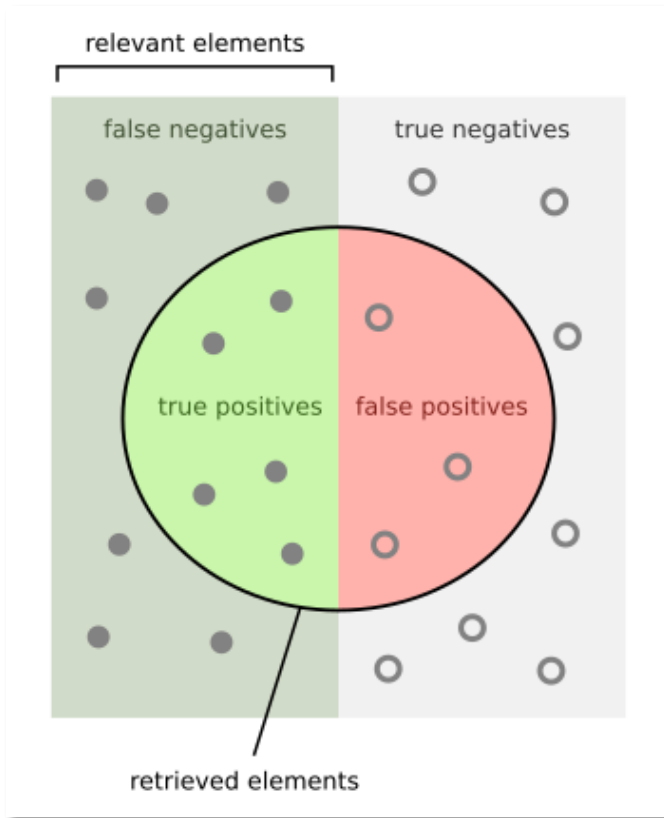
Think about **onboarding**:

- How do you get relevant initial information about new users?
- And how does this impact your UI?
E.g. an onboarding view on first launch



Precision vs Recall

As a metric for evaluating recommendation algorithms



$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

„What fraction of the retrieved & shown recommendations is actually relevant to the user?“

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

„What fraction of all items that would be relevant do we actually retrieve & show to the user?“

Image from: https://en.wikipedia.org/wiki/Precision_and_recall

Precision vs Recall

As a UI design factor and tradeoff

Note: The illustrated „conversation“ is a key point of this slide.
Precision vs recall can be discussed as an **explicit design factor**,
you don't have to just „let it happen“!

We should go for **high precision**!
Let's try to **not show irrelevant items**
to the user!

⚠ **Risk:**
Miss some
relevant options



We should go for **high recall**!
Let's try to show **all relevant items**
to the user!

⚠ **Risk:**
Show many
irrelevant options



Favouring Precision or Recall

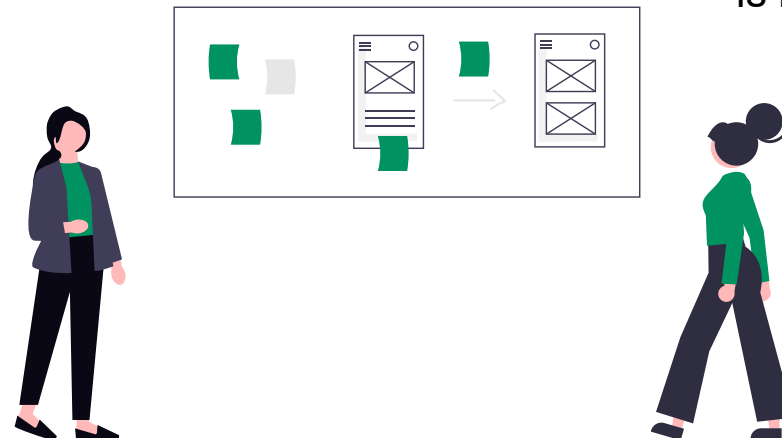
Indicators that help decide in a specific design context / use-case

Consider favouring high precision when

- Users want to decide **quickly**, so we only want to give them good options
- The cost of engaging with an irrelevant option is high
- The cost of missing a (specific) relevant option is low

Consider favouring high recall when

- Users want to **explore**, so we can show many options
- The cost of engaging with an irrelevant option is low
- The cost of missing a (specific) relevant option is high



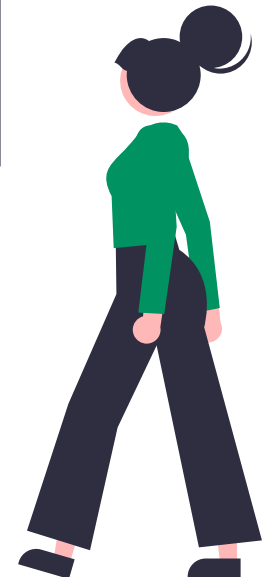
Activity

Design a UI with recommendations

Adjust your UI sketch or sketch a new iteration.

Think about **precision vs recall**:

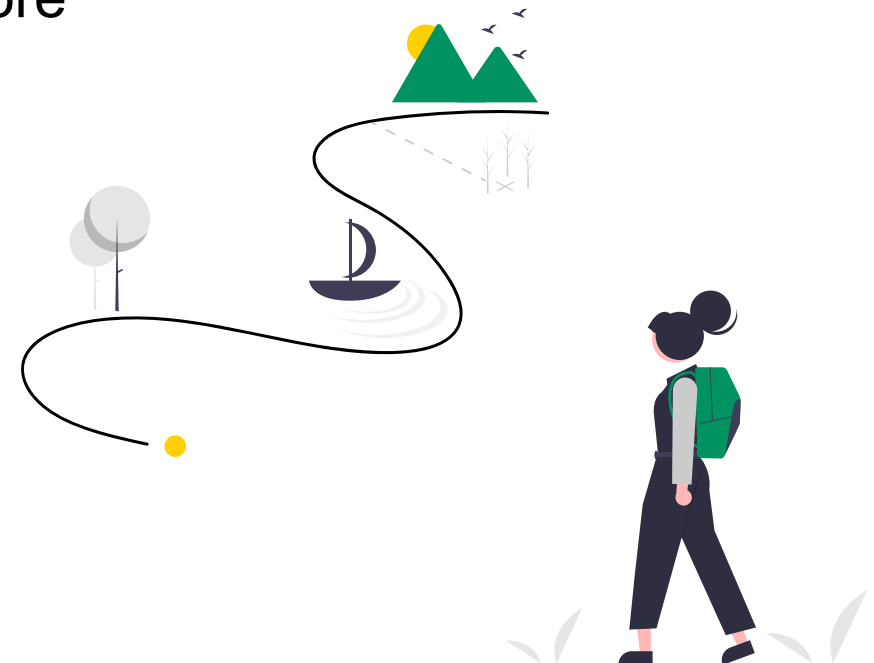
- Does your use-case have a reason to favour precision or recall? Why?
- Does this impact your UI?



Exploitation vs Exploration

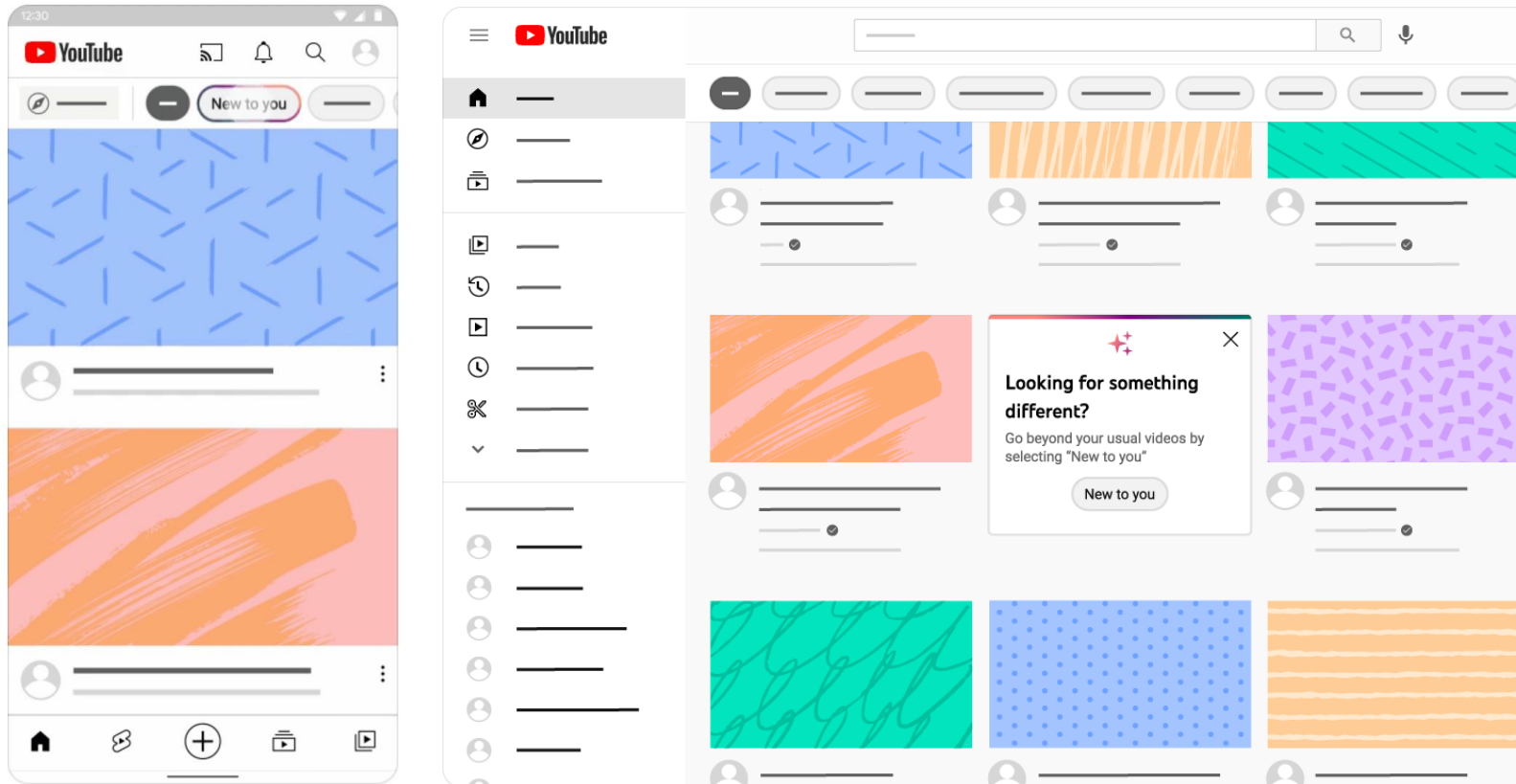
„Safe bets“ vs discovery

- Typically, “recommendation” implies giving the user more of what they have seen / used / liked before
- Alternative: Designing for **exploration & discovery!**



UI features for exploration

Product example: „New to you“ features in YouTube

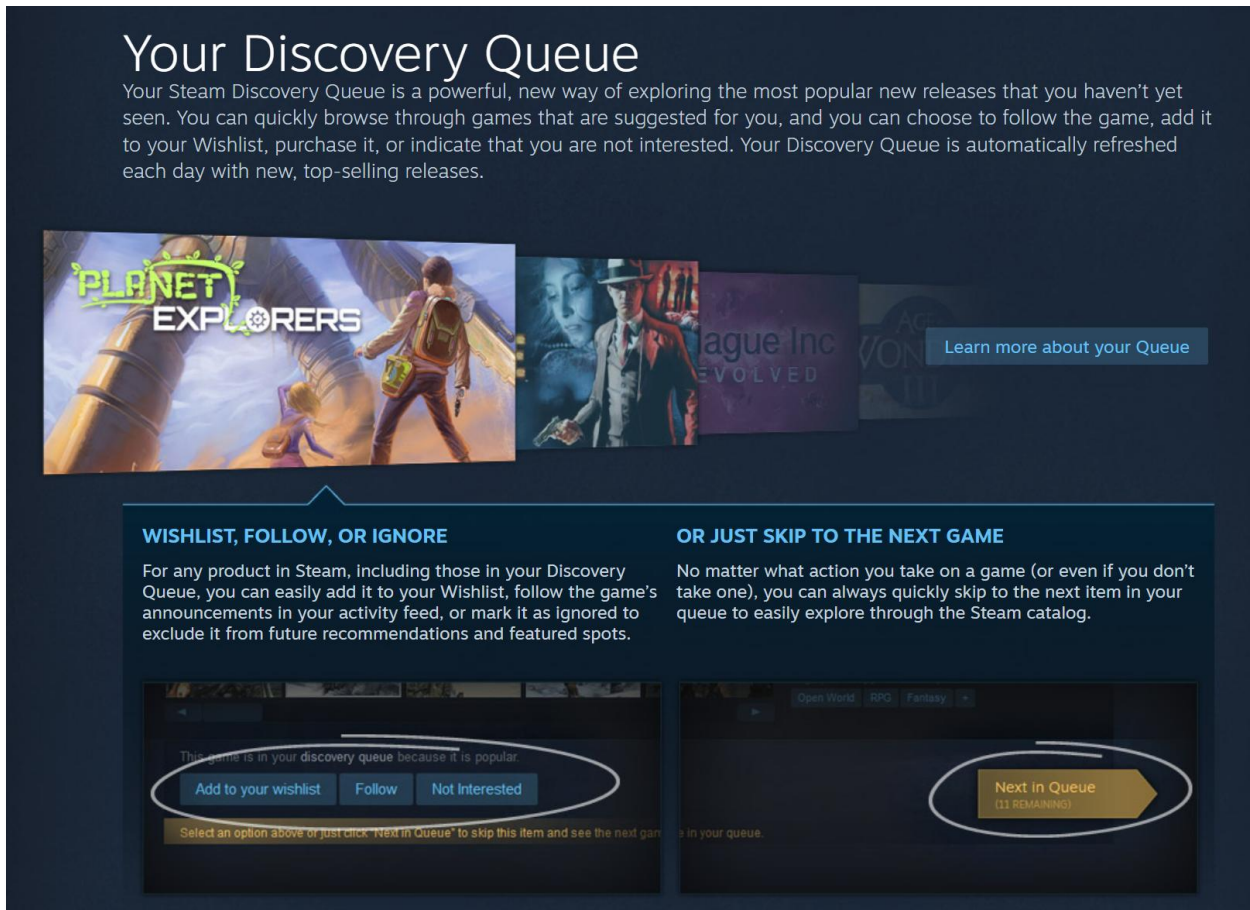


“a feature helping you **discover** new creators and fresh content, **beyond the recommended** videos you normally see”

<https://support.google.com/youtube/thread/132416851/new-to-you-discover-more-content-on-the-youtube-homepage>

UI features for exploration

Product example: „Discovery queue“ on Steam



Notes:

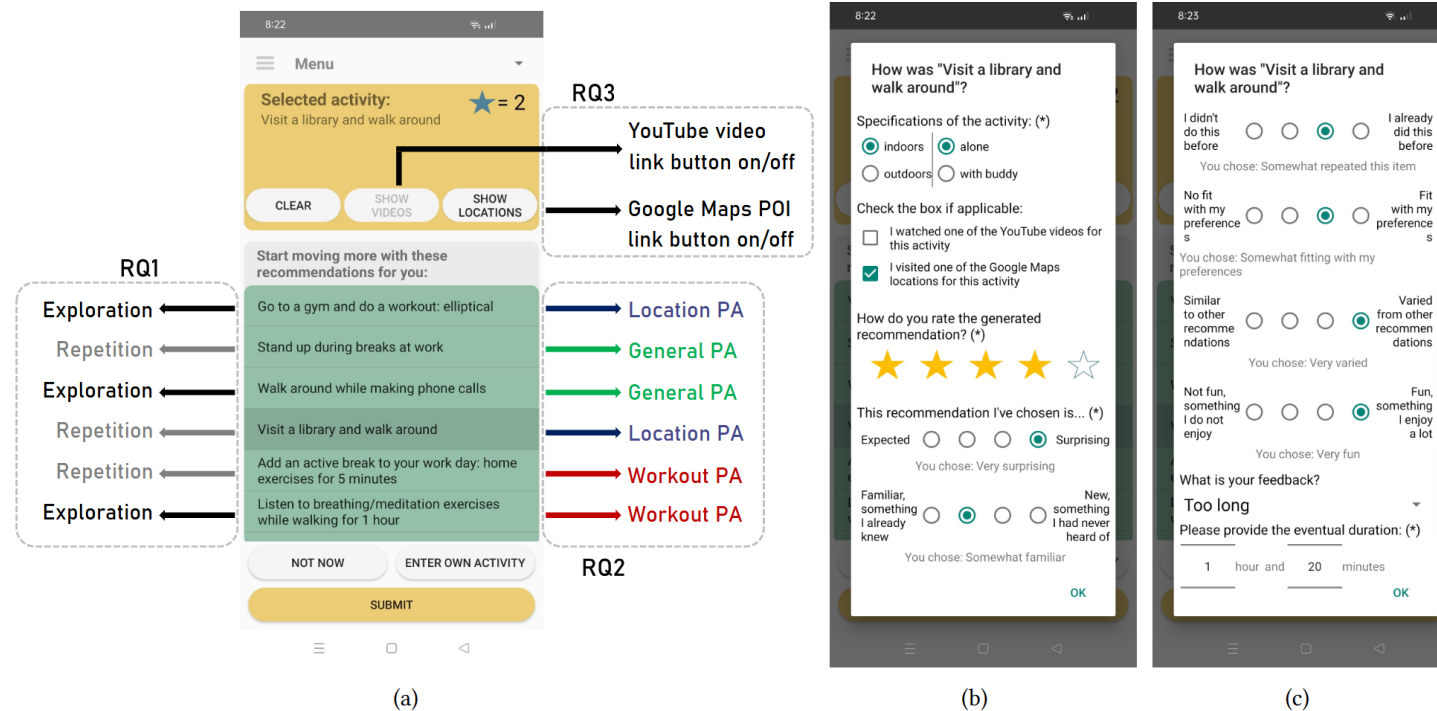
It's not known to what extent this queue actually tries to suggest games that deviate from the user's taste but there are a few hints:

- The text says the queue includes „most popular new releases“ (instead of the best fitting ones for the individual user)
- The UI design and even this explanation text highlight fast skipping („or just skip“), which makes sense if this UI feature is designed with the risk of showing irrelevant games in mind

<https://store.steampowered.com/about/newstore>

Comparing exploitation & exploration

Research example: Study with an app that recommends physical activities



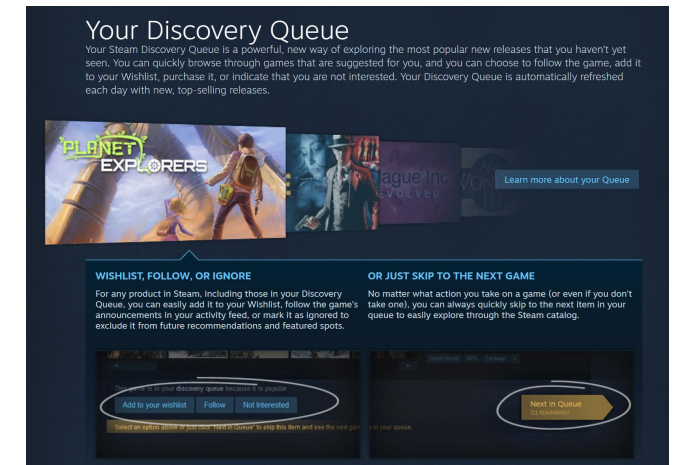
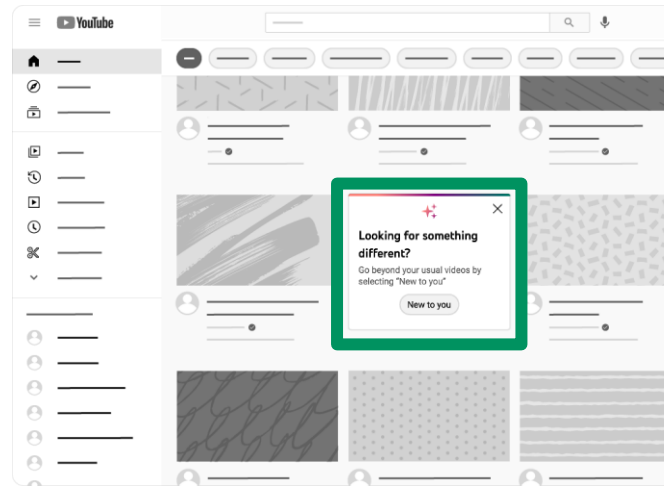
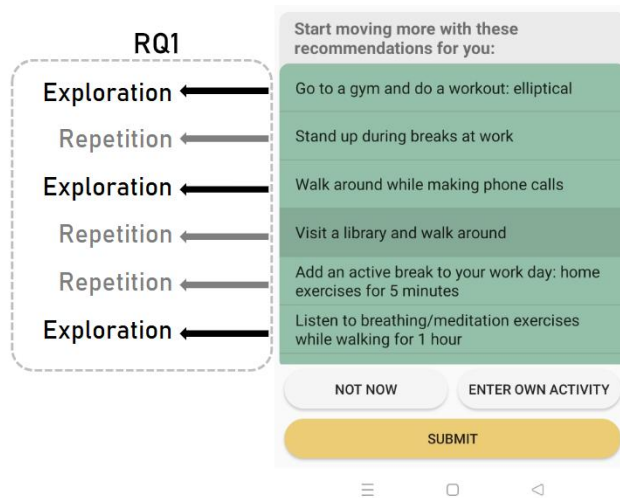
“Our results suggest that **physically inactive adults prefer exploration** of [physical activities], as these were submitted the most and resulted in significantly higher star ratings and durations.”

Figure 2: In the main screen of the app, three types of randomizations can be distinguished, corresponding to each of the research questions (a). When a user wants to submit a PA recommendation item, feedback on the location, company, POI visit/video watching, star rating, manipulation check variables, and eventual duration are requested in a feedback screen (b and c).

[Coppens et al., 2024]

Integrating exploration

UI design choices, using our examples



Integrated

Exploration-focused items appear in the same way as all recommendations

Mixed

Hints at exploration integrated with the usual recommendations

Separate

Exploration-focused items appear in a separate UI (e.g. a dedicated mode)

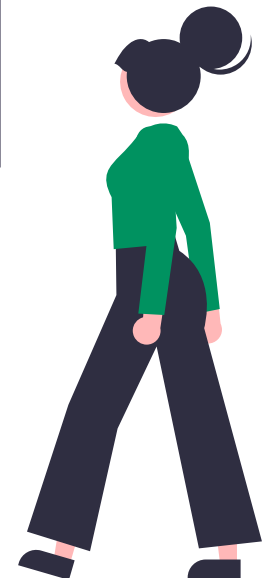
Activity

Design a UI with recommendations

Adjust your UI sketch or sketch a new iteration.

Think about **exploration**:

- Should your UI support your target group explicitly in exploring „new directions“?
- If so, how can you integrate that into your UI?



Explaining recommendations

Typical types of explanations; they differ based on what they refer to:

„You have been watching a lot of sports.
This is the **most popular** world club clip
right now.“

Top item

1.

- Relate to the top item in a context or in general
- E.g. most liked of a genre / overall

„You have been watching a lot of sports.
This is **in the most liked** clips right now.“

Top-N items

1.

2.

...

- Relate to the top N items in a context or in general
- E.g. among most liked of a genre

See: <https://ieeexplore.ieee.org/document/4401070>

Explaining recommendations

Typical types of explanations; they differ based on what they refer to:

„Since you liked X [from Y], you might also like this [from Y]: ...“

Similar to item



- Relate item to another one
- Possibly mention a **dimension of similarity**

“People **like you** liked this: ...”

“Others **who liked rock**, also liked this: ...”

Similar to users



- Relate item to users, referring to **(user-based) similarity**
- Could also mention a concrete **dimension of similarity**

See: <https://ieeexplore.ieee.org/document/4401070>

Scoring recommendations

Another approach: Show a ranking/rating/score/prediction value for each item



Rank

1.

2.

3.

4.

**Numerical score
or probability**

98 %

77 %

56 %

24 %

Verbal score

Very high

High

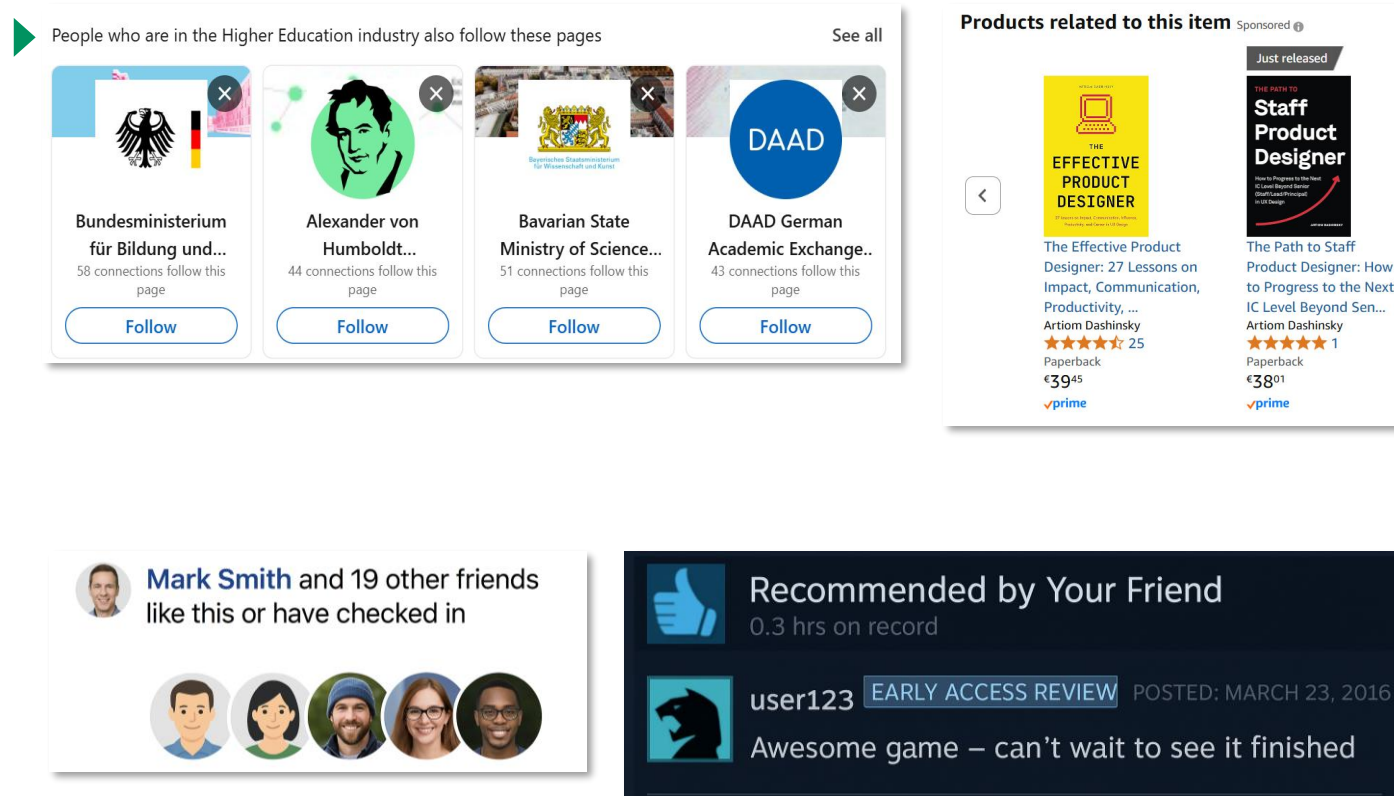
Medium

Low

Q: What does this „explain“? What insight can users gain from such scores?

“Social signals”

UI elements that link recommendations to other people



Abstract link

- Mentions common aspect / „community“
- Related: Shows how others have rated in aggregation (e.g. average stars)

Concrete link

- Mentions one or more specific people

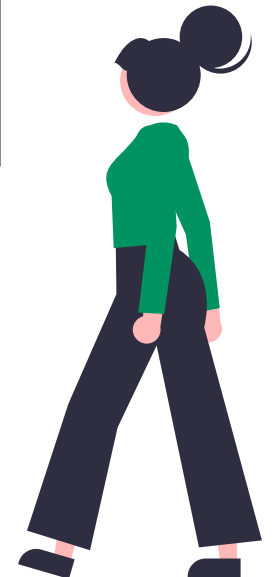
Activity

Design a UI with recommendations

Adjust your UI sketch or sketch a new iteration.

Think about **explanations** and “**social signals**”:

- Should your UI explain recommendations and/or provide references to other users?
- If so,
 - how can you integrate that into your UI?
 - in a way that is adequate for your target group?

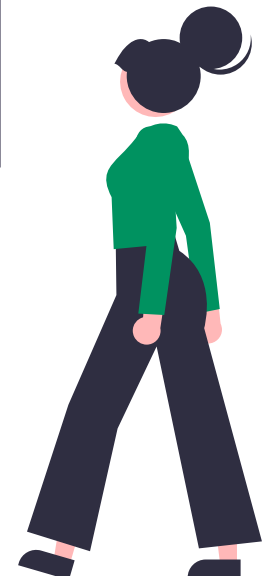


Well done! 🙌

You've designed a UI with recommendations, considering many relevant design dimensions!

Feel free to share your UI and reflections with the others!

And/or take a picture and email it to:
daniel.buschek@uni-bayreuth.de

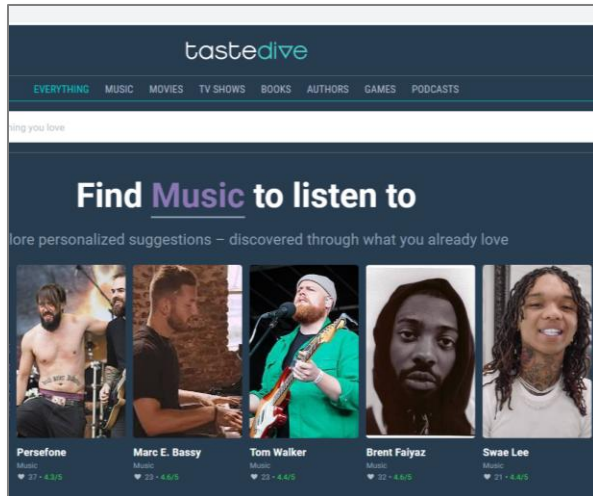


Outlook

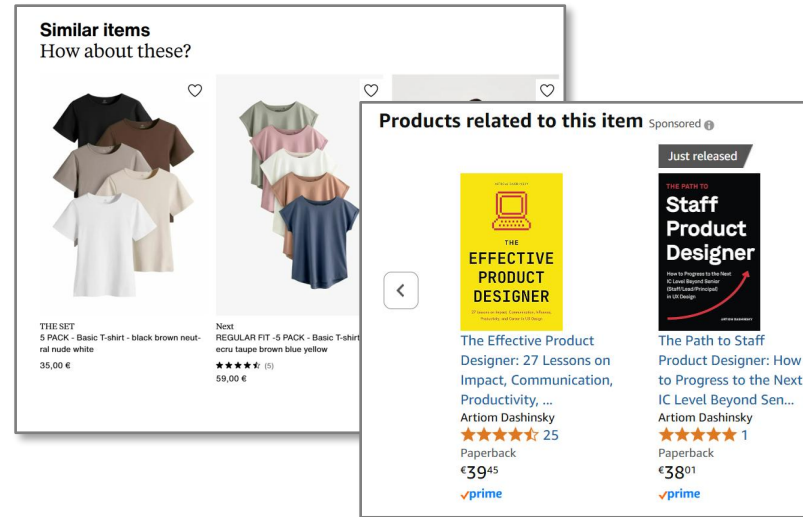
Further design aspects and directions

Who takes initiative in interaction with recommendations?

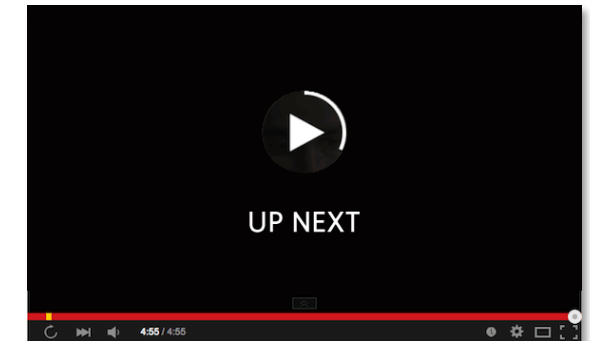
User or system, or mixed? Some example arguments



User initiative,
e.g. by searching/browsing
on a dedicated music
recommendation website



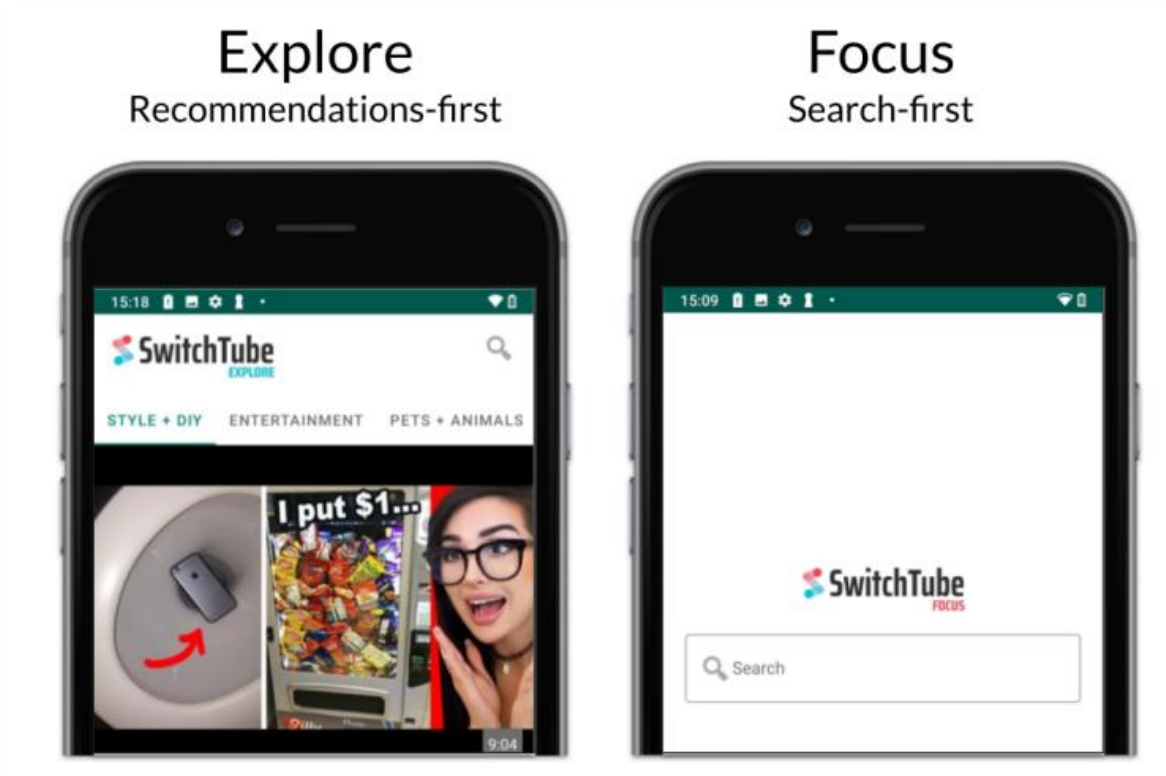
Mixed?
E.g. integrated recommendations
in websites for use-cases where users
have a typical interest in browsing more



System initiative,
e.g. by directly choosing
and playing a next video

Who has control over recommendations?

A recent study



SwitchTube: A Proof-of-Concept System
Introducing “Adaptable Commitment Interfaces”
as a Tool for Digital Wellbeing
[Lukoff et al., 2023]

Who has control?

Giving users „control over control“

“In a three-week field deployment with 46 US participants, we evaluate how the **ability to switch between interfaces** affects user experience, finding that it provides users with a greater sense of **agency**, **satisfaction**, and **goal alignment**.”

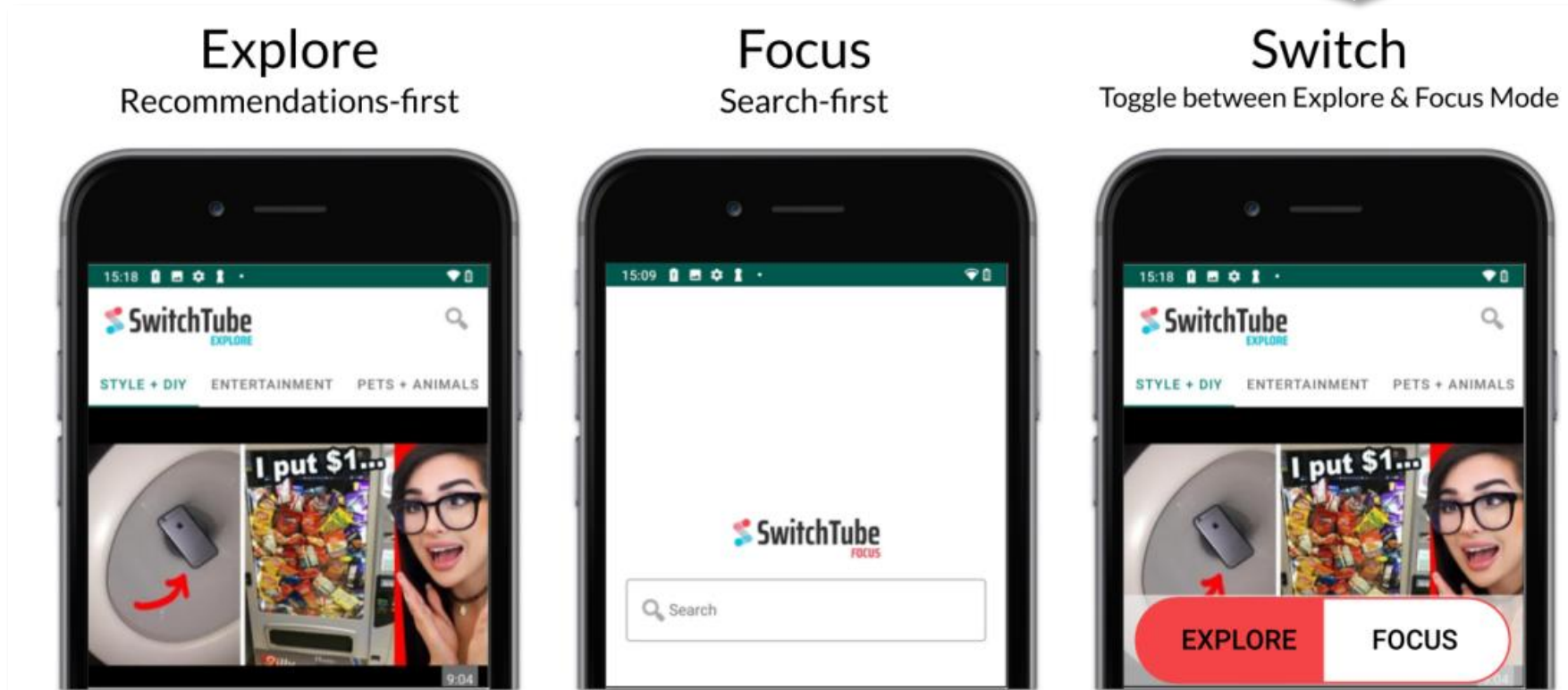


Figure 1: The three different versions of the SwitchTube study app are designed to support different levels of sense of agency for the user. The Explore Version (low sense of agency) has a recommendations-first homepage whereas the Focus Version (medium) has a search-first homepage. The Switch Version (high) is an “adaptable commitment interface” that lets users toggle between Explore and Focus Mode.

[Lukoff et al., 2023]

“Forward” vs “backward” reasoning

Supporting people in forward reasoning, not just recommending options to think about

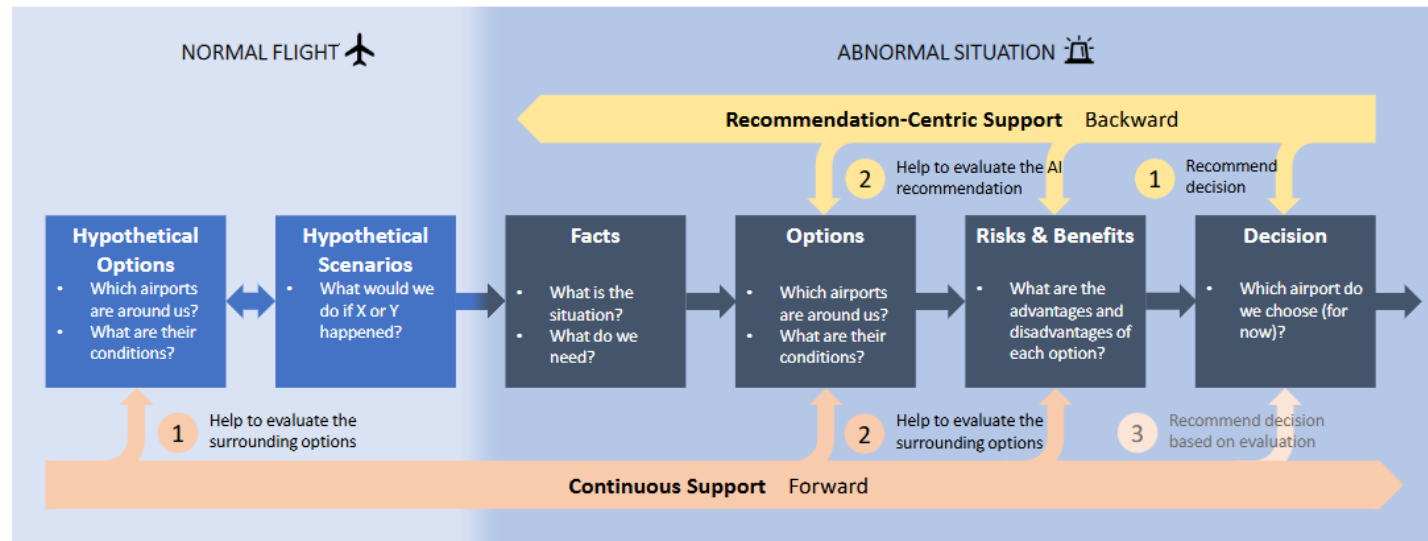


Fig. 1. Conceptual overview of the two decision support paradigms that we compare: *recommendation-centric support* and *continuous support*. While the former pushes pilots to reason backward from a decision recommendation, the aim of the latter is to support pilots in a forward direction. Instead of only popping up during an emergency, the system continuously helps pilots to evaluate their surroundings even during normal flight. The system does not give recommendations to avoid biasing pilots. But given continuous support, it may be possible to add recommendations in an emergency while still allowing pilots to reason forward, since pilots are already engaged with the system when they see the recommendations.

Beyond Recommendations:
From Backward to Forward AI
Support of Pilots' Decision-
Making Process
[\[Zhang et al., 2024\]](#)

Further aspects & reading

- **Group recommendations**
 - Recommendations need to fit to the interests of **multiple** people
 - Might involve „negotiation“ among users, e.g. via voting, chat, etc.
 - See: [*A Systematic Review of Interaction Design Strategies for Group Recommendation Systems*](#)
- **User perspective:** What do people think who have not studied CS / HCI / AI?
 - See: [*Middle-Aged Video Consumers' Beliefs About Algorithmic Recommendations on YouTube*](#)
- **Meta:** History & context on the human-centered perspective on recommendations (as opposed to focusing on the algorithms in research)
 - See: [*Human-centered recommender systems: Origins, advances, challenges, and opportunities*](#)

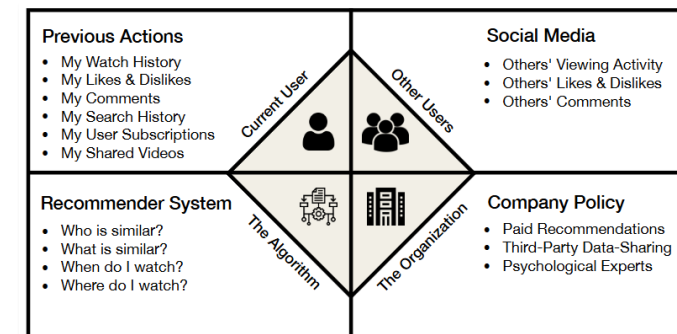
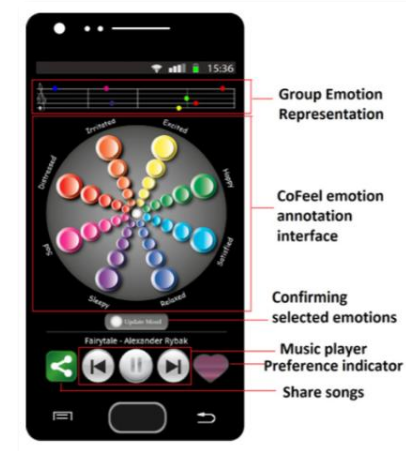
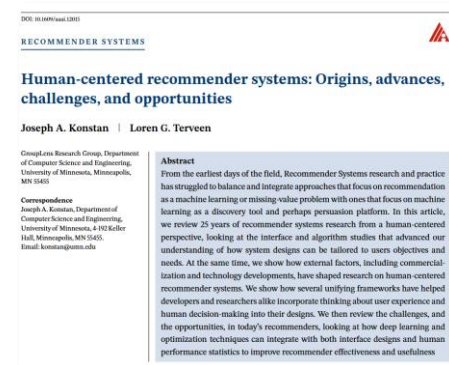


Fig. 1. The Four Actors that Influence Video Recommendations and User Beliefs on YouTube.



References

- Oscar Alvarado, Hendrik Heuer, Vero Vanden Abeele, Andreas Breiter, and Katrien Verbert. 2020. Middle-Aged Video Consumers' Beliefs About Algorithmic Recommendations on YouTube. *Proc. ACM Hum.-Comput. Interact.* 4, CSCW2, Article 121 (October 2020), 24 pages. <https://doi.org/10.1145/3415192>
- Ine Coppens, Toon De Pessemier, and Luc Martens. 2024. Repeating my Workouts or Exploring new Activities? A Longitudinal Micro-Randomized User Study for Physical Activity Recommender Systems. In Adjunct Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization (UMAP Adjunct '24). Association for Computing Machinery, New York, NY, USA, 176–182. <https://doi.org/10.1145/3631700.3664867>
- Kai Lukoff, Ulrik Lyngs, Karina Shirokova, Raveena Rao, Larry Tian, Himanshu Zade, Sean A. Munson, and Alexis Hiniker. 2023. SwitchTube: A Proof-of-Concept System Introducing “Adaptable Commitment Interfaces” as a Tool for Digital Wellbeing. In Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 197, 1–22. <https://doi.org/10.1145/3544548.3580703>
- Zelun Tony Zhang, Sebastian S. Feger, Lucas Dullenkopf, Rulu Liao, Lou Süsslin, Yuanting Liu, and Andreas Butz. 2024. Beyond Recommendations: From Backward to Forward AI Support of Pilots' Decision-Making Process. *Proc. ACM Hum.-Comput. Interact.* 8, CSCW2, Article 485 (November 2024), 32 pages. <https://doi.org/10.1145/3687024>